
Minimax Optimal Strategy for Delayed Observations in Online Reinforcement Learning

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Abstract

We study reinforcement learning with delayed state observation, where the agent observes the current state after some random number of time steps. We propose an algorithm that combines the augmentation method and the upper confidence bound approach. For tabular Markov decision processes (MDPs), we derive a regret bound of $\tilde{O}(H\sqrt{D_{\max}SAK})$, where S and A are the cardinalities of the state and action spaces, H is the time horizon, K is the number of episodes, and $D_{\max}$ is the maximum length of the delay. We also provide a matching lower bound up to logarithmic factors, showing the optimality of our approach. Our analytical framework formulates this problem as a special case of a broader class of MDPs, where their transition dynamics decompose into a known component and an unknown but structured component. We establish general results for this abstract setting, which may be of independent interest.

1. Introduction

Reinforcement learning (RL) is a framework where machines learn to make optimal decisions under the guidance of rewards (Sutton & Barto, 2018). The learning process involves interaction between the agent and the environment. The agent observes the state of the environment and makes an action, which affects the reward the agent receives and the transition probability of the next state. While there have been extensive theoretical advances and empirical successes in RL, one barrier to applying RL to real-world tasks is the delayed accessibility of the current state. For instance, domains like robotics and autonomous driving encounter inevitable delays in observations due to sensor processing,

data transmission, or computational overhead. In online advertising, a sequence of ads must be planned without observing the users’ internal orientation toward the products. This is problematic because standard methods in RL assume and heavily rely on observing the current state of the environment. With delays in state observation, the agent is forced to plan a sequence of actions in advance. As the number of possible action sequences increases exponentially with the length of the delay, even a minor delay may substantially complicate the problem.

While various empirical methods have been proposed to overcome this challenge (Agarwal & Aggarwal, 2021; Kim et al., 2023; Wu et al., 2024; 2025), there is currently a large gap in the theoretical understanding of this setting. Walsh et al. (2009); Chen et al. (2023) show that obtaining polynomial sample complexity is possible despite the exponential growth of action sequences, but their bounds appear quite loose, and it is not clear what the optimal dependency is. Moreover, it has not yet been established whether the length of the delay should affect the sample complexity. In this work, we close these gaps by providing matching upper and lower regret bounds for this setup, providing a better understanding of learning MDPs with delayed observations.

We summarize our main contributions as follows:

- We propose an algorithm for learning delayed MDPs that constructs an equivalent augmented MDP without delays and then applies an upper confidence bound (UCB) method. Several key distinctions in the augmentation process from previous works enable a better algorithmic design and a clear explanation of the theoretical guarantees.
- Under the tabular MDP setting, we derive a regret bound of $\tilde{O}(H\sqrt{D_{\max}SAK})$. This result improves upon the previously best known bound by Chen et al. (2023) by a factor of $H^{1/2}D_{\max}^2$.
- We provide a matching lower bound of $\tilde{\Omega}(H\sqrt{D_{\max}SAK})$, highlighting the optimality of our algorithm and analysis. To the best of our knowledge, this is the first result that rigorously shows that shorter delays (relative to the horizon H) decrease the statistical complexity of the problem.

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- We identify the core properties of delayed MDPs and abstract them to define a more general setting of MDPs whose transition dynamics decompose into a fully known part and an unknown but structured part. We provide theoretical results for this setting, which naturally extends to the delayed observation setting as a special case. We emphasize that our results apply to any problem domain that fits this abstraction.

1.1. Related Work

RL under Tabular MDPs. A variety of theoretically guaranteed methods for learning tabular MDPs have been developed and analyzed (Azar et al., 2017; Jin et al., 2018; Zanette & Brunskill, 2019; Dann et al., 2019; Zhang et al., 2021; Tiapkin et al., 2022; Zhang et al., 2024; Lee & Oh, 2025). These works achieve the minimax optimal regret bound of $\tilde{O}(H\sqrt{SAK})$ for the time-homogeneous setting or $\tilde{O}(H^{3/2}\sqrt{SAK})$ for the time-inhomogeneous setting, where the lower bound is provided by Jaksch et al. (2010); Domingues et al. (2021).

MDPs with Delayed Observations. Attention on MDPs with delayed observations dates back more than twenty years (Bander & White III, 1999; Katsikopoulos & Engelbrecht, 2003; Walsh et al., 2009), with a heavier focus on characterizing the properties of delayed MDPs, such as defining the equivalent augmented MDP and computational hardness, rather than learning optimal policies. Walsh et al. (2009) describe a polynomial (ε, δ) -PAC bound of $\tilde{O}(\frac{S^2A}{\varepsilon^3(1-\gamma)^6})$ for γ -discounted MDPs with constant delay. A recent work by Chen et al. (2023) study the learning problem of delayed MDPs using modern techniques. They assume stochastic delays with unknown distributions, imposing that the delay distribution must be learned along with the transition distribution. They achieve a regret bound of $\tilde{O}(H^{3/2}D_{\max}^{5/2}\sqrt{SAK})$ for the time-homogeneous finite-horizon setting. They also provide a regret lower bound of $\Omega(\sqrt{HD_{\max}SAK})$, leaving a gap of $\tilde{O}(HD_{\max}^2)$ between the two bounds.

Delayed feedback in bandits and RL. A related yet distinct line of work studies efficient learning under delayed feedback in bandits (Zhou et al., 2019; Vernade et al., 2020; Gael et al., 2020; Lancewicki et al., 2021; Masoudian et al., 2022; Howson et al., 2023a) and RL (Howson et al., 2023b; Lancewicki et al., 2023; Mondal & Aggarwal, 2023; Kuang et al., 2023; Yang et al., 2023). These works assume that the data generated by a policy reach the learner with a delay, but the executor of the policy has access to the current state. This fact constitutes a clear distinction from our setting, where the delay occurs in the execution process.

2. Preliminaries

2.1. Notations

For a measurable set $\mathcal{Z}$, we denote the set of all probability distributions over $\mathcal{Z}$ by $\Delta(\mathcal{Z})$. For a function of the form $P : \mathcal{X} \rightarrow \Delta(\mathcal{Z})$, we frequently write P_x as shorthand notation for $P(x)$. For a distribution $P \in \Delta(\mathcal{Z})$ and a function $V : \mathcal{Z} \rightarrow \mathbb{R}$ over the same space, we denote the expectation of V under P by $PV := \sum_{z \in \mathcal{Z}} P(z)V(z)$, and the variance of V under P by $\mathbb{V}(P, V) := \sum_{z \in \mathcal{Z}} P(z)(V(z) - PV)^2$.

For two integers $m \leq n$, let $[m : n] := \{m, \dots, n\}$, and if $n \geq 1$, we define $[n] := [1 : n]$. For real numbers x, a, b , let $a \wedge b := \min\{a, b\}$ and $\text{clip}(x; a, b) := \min\{\max\{x, a\}, b\}$.

We frequently deal with a queue of actions, which we denote by $\mathbf{a} = (a_i, a_{i+1}, \dots, a_j) \in \cup_n \mathcal{A}^n$. For an action queue $\mathbf{a} = (a_1, a_2, \dots, a_n) \in \mathcal{A}^n$, we denote the first element by $\mathbf{a}_1 := a_1$, the queue starting from the second element by $\mathbf{a}_2 := (a_2, \dots, a_n)$, its length by $\text{len}(\mathbf{a}) := n$, and concatenation by $\mathbf{a} + a := (a_1, \dots, a_n, a)$ for $a \in \mathcal{A}$.

2.2. Markov Decision Process

We consider a finite-horizon episodic MDP $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, r, H)$ with a time-homogeneous transition kernel. $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, $P : \mathcal{S} \times \mathcal{A} \rightarrow \Delta(\mathcal{S})$ is the transition kernel, $r : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ is the reward function, and H is the time horizon of an episode. We assume that the state and action spaces have finite cardinalities S and A , respectively. We also assume that the reward function satisfies $r(s, a) \in [0, 1]$ and is known to the agent¹. An agent interacts with the MDP for K episodes. At the k -th episode, through time steps $h = 1, \dots, H$, the agent observes a state $s_h^k \in \mathcal{S}$, takes an action $a_h^k \in \mathcal{A}$, receives a reward $r(s_h^k, a_h^k)$, and the next state is sampled as $s_{h+1}^k \sim P(s_h^k, a_h^k)$. A policy $\pi = \{\pi_h\}_{h=1}^H$ is a sequence of functions $\pi_h : \mathcal{S} \rightarrow \mathcal{A}$ that determines the action given the current state. The value function of a policy is defined as $V_h^\pi(s) := \mathbb{E}_{\pi(\cdot|s_h=s)}[\sum_{j=h}^H r(s_j, a_j)]$. The optimal policy is defined as the policy π^* that satisfies $V_h^{\pi^*}(s) = V_h^*(s) := \max_{\pi} V_h^\pi(s)$ for all h and s . The goal of the agent is to minimize the cumulative regret over K episodes, which is defined as $\sum_{k=1}^K (V_1^*(s_1^k) - V_1^{\pi^k}(s_1^k))$.

We define the *branching factor* B as an upper bound on the support sizes of $P(s, a)$, that is, the number of candidate states one can reach by taking the state-action pair. Formally, we have $|\{s' \in \mathcal{S} \mid P_{s,a}(s') > 0\}| \leq B \leq S$ for all

¹The analysis in this paper remains valid when the assumption $r(s, a) \in [0, 1]$ is relaxed to $0 \leq V_h^\pi(s) \leq H$ for all h, s , and π , which is a generalization noted in Lee & Oh (2025). Also, an unknown reward function can be learned by standard methods (Zanette & Brunskill, 2019; Zhang et al., 2024), which are applicable to the delayed observation setting.

(s, a) . We assume B is known to the agent, which is always possible since the agent may set $B = S$.

3. Problem Setting

We consider stochastic delayed MDPs (SDMDPs) (Katsikopoulos & Engelbrecht, 2003), which are MDPs with delayed state observation. An SDMDP consists of an MDP $\mathcal{M}$, a delay distribution $P_{\text{delay}} : \mathcal{S} \times \mathcal{A} \rightarrow \triangle([-1 : \Delta_{\max}])$, and an upper bound on the delay length $D_{\max}$. The main difference in this setting is that the current state is revealed to the agent after some time steps. As a result, the agent must take an action while the current state is still hidden, then observes the effect of the action after the delay. Specifically, after taking the h -th action a_h , the following next state s_{h+1} is revealed to the agent after D_h time steps, that is, at the beginning of the $h + 1 + D_h$ -th time step. If $h + 1 + D_h \geq H + 1$, then the state is revealed after the end of the episode. We assume that the delay D_h is determined in the following way.

Assumption 1. *For each time step h , an inter-arrival time Δ_h is sampled from the distribution $P_{\text{delay}}(s_h, a_h)$ independently of all other randomness including the sampling of s_{h+1} . Starting from $D_0 = 0$, which implies that the initial state s_1 is always immediately observed, the delay D_h is determined incrementally as $D_h = \text{clip}(D_{h-1} + \Delta_h; 0, D_{\max})$, where $D_{\max}$ is a known constant that upper-bounds D_h .*

Remark 1. In the previous work (Katsikopoulos & Engelbrecht, 2003; Chen et al., 2023), the inter-arrival time Δ_h is assumed to be distributed over $[0 : \Delta_{\max}]$, implying that at most one state is revealed per time step. Generalizing their assumption, we allow an inter-arrival time of -1 , in which case multiple consequent states are revealed at the same time step.

Constant Delayed MDP (CDMDP). The case where the length of the delay D_h is equal to $D_{\max}$ for all $h \in [H]$ is called the *constant delayed MDP (CDMDP)* (Walsh et al., 2009). Assumption 1 recovers this setting by adding an auxiliary initial state s_{start} with a fixed inter-arrival time $D_{\max}$ and setting all the other inter-arrival times to 0, making the total length of the delay $D_h = \sum_{j=1}^h \Delta_j$ equal to $D_{\max}$.

We consider both cases of SDMDPs where P_{delay} is known and P_{delay} is unknown. When unknown, following Chen et al. (2023), we allow the agent to learn the distribution by assuming that the original values of $\Delta_1, \dots, \Delta_H$ are revealed at the end of the episode, which is necessary since their values may be lost when the delay is clipped or the episode terminates. We emphasize that access to Δ_h during the episode is not allowed.

With observational delays, the policy can no longer be a

Table 1. Example trajectory of a CDMDP with $H = 6$ and constant delay 2. Each column denotes the time step h , the last observed state s_{t_h} , the action queue $\mathbf{a}$, the inter-arrival time counter $\tilde{\Delta}_h$, the inter-arrival time Δ_{t_h} , and the total delay D_{t_h} . Note that a new state is observed when $\tilde{\Delta}_h$ is equal to Δ_{t_h} , or equivalently, when the length of the queue $\text{len}(\mathbf{a})$ is equal to D_{t_h} .

h	s_{t_h}	$\mathbf{a}$	$\tilde{\Delta}_h$	Δ_{t_h}	D_{t_h}
1	s_1	$\emptyset$	0	2	2
2	s_1	(a_1)	1	-	-
3	s_1	(a_1, a_2)	2	-	-
4	s_2	(a_2, a_3)	0	0	2
5	s_3	(a_3, a_4)	0	0	2
6	s_4	(a_4, a_5)	0	0	2
End	s_7	$\emptyset$	-	-	-

Table 2. Example trajectory of an SDMDP with $H = 6$ with one possible realization of delays. Refer to Table 1 for the description of each column. States s_3 and s_4 are revealed at the same time step $h = 6$ because $\Delta_3 = -1$.

h	s_{t_h}	$\mathbf{a}$	$\tilde{\Delta}_h$	Δ_{t_h}	D_{t_h}
1	s_1	$\emptyset$	0	1	1
2	s_1	(a_1)	1	-	-
3	s_2	(a_2)	0	2	3
4	s_2	(a_2, a_3)	1	-	-
5	s_2	(a_2, a_3, a_4)	2	-	-
6	s_3	(a_3, a_4, a_5)	(-1)	-1	2
	s_4	(a_4, a_5)	0	1	3
End	s_7	$\emptyset$	-	-	-

mapping from a state to an action since the current state is not observable. Instead, the policy must consider three elements. First, we let t_h be the time index of the last state observable at time step h . If a state is revealed at time h , it is state s_{t_h} , otherwise we have $t_h = t_{h-1}$. Next, let $\mathbf{a} = (a_{t_h}, a_{t_h+1}, \dots, a_{h-1})$ be the queue of unresolved actions. It is straightforward that the policy must consider s_{t_h} and $\mathbf{a}$. In addition, the policy must take the time step that revealed s_{t_h} into account, since it affects when the next state s_{t_h+1} is revealed under Assumption 1. We let $\tilde{\Delta}_h = h - (t_h + D_{t_h-1})$ be the number of time steps elapsed since a state was revealed; recall that $t_h + D_{t_h-1}$ is the time step at which s_{t_h} is revealed. For example, in the constant $D_{\max}$ delay setting, we have $\tilde{\Delta}_1 = 0, \tilde{\Delta}_2 = 1, \tilde{\Delta}_3 = 2, \dots, \tilde{\Delta}_{D_{\max}+1} = D_{\max}$, and $\tilde{\Delta}_{D_{\max}+i} = 0$ for all $i > 1$ since we observe a new state at each time step, just delayed by $D_{\max}$. It can be shown that the information of these three elements is sufficient to make optimal actions in an SDMDP (Katsikopoulos & Engelbrecht, 2003). Consider the following policy class:

$$\Pi_{\text{delay}} = \left\{ \{\pi_h\}_{h=1}^H : \mathcal{S} \times \cup_{D=0}^{D_{\max}} \mathcal{A}^D \times [0 : \Delta_{\max}] \rightarrow \mathcal{A} \right\}.$$

A policy $\pi \in \Pi_{\text{delay}}$ chooses an action $a_h = \pi_h(s_{t_h}, \mathbf{a}, \tilde{\Delta})$ at time step h . For a given SDMDP $(\mathcal{M}, P_{\text{delay}}, D_{\max})$, we can also define the value function of a policy as $V_1^\pi(s) = \mathbb{E}_{\pi(\cdot|s_1=s)}[\sum_{h=1}^H r(s_h, a_h)]$, and the optimal policy π_{delay}^* that satisfies $V_1^{\pi_{\text{delay}}^*}(s) = \max_{\pi \in \Pi_{\text{delay}}} V_1^\pi(s)$ for all $s \in \mathcal{S}$. The cumulative regret that the agent must minimize is redefined as $\sum_{k=1}^K (V_1^{\pi_{\text{delay}}^*}(s_1^k) - V_1^{\pi^k}(s_1^k))$.

Connection to Partially Observable MDPs (POMDPs).

POMDPs are a framework in which latent states emit random observations rather than being directly observed (Papadimitriou & Tsitsiklis, 1987; Jin et al., 2020). Having limited observability, the delayed observation model may be viewed as a special case of this framework. Specifically, an SDMDP can be modeled as a POMDP by treating a sequence of the last observed state followed by the unobserved states as a single latent state, with only the first state in the sequence being observable. One benign condition for POMDPs that may have a connection to the delayed MDPs is *multi-step revealing*, which posits that the latent state can be statistically inferred from a number of following observations (Liu et al., 2022; 2023; Zhang & Jiang, 2025). While the delayed MDPs fall into this subclass of POMDPs, techniques for this setting incur regret exponential in the delay length. This fact indicates that reduction to generic POMDPs fails to exploit the structure induced by the delays. See Walsh et al. (2009); Chen et al. (2023) for further discussion on why methods for POMDPs are not applicable to delayed MDPs.

4. Algorithm

In this section, we describe our algorithm for learning SDMDPs. There are two main steps. First, we construct an augmented MDP that is equivalent to the given SDMDP. Then, we learn the augmented MDP by applying standard techniques with certain modifications that exploit the properties of the augmented MDP. Each step is explained in the following subsections.

4.1. Augmented Markov Decision Process

Given an SDMDP $(\mathcal{M}, P_{\text{delay}}, D_{\max})$, we construct the following equivalent augmented MDP (without delay) $\mathcal{M}_{\text{aug}} = (\mathcal{S}_{\text{aug}}, \mathcal{A}, P_{\text{aug}}, r_{\text{aug}}, H)$. The intuition is to treat the tuple of the last observed state s_{t_h} , the action queue $\mathbf{a}_h = (a_{t_h}, \dots, a_{h-1})$, and the number of time steps without new observation $\tilde{\Delta}_h$ as an augmented state. We note that this construction is, in principle, equivalent to the ones in Katsikopoulos & Engelbrecht (2003); Chen et al. (2023),

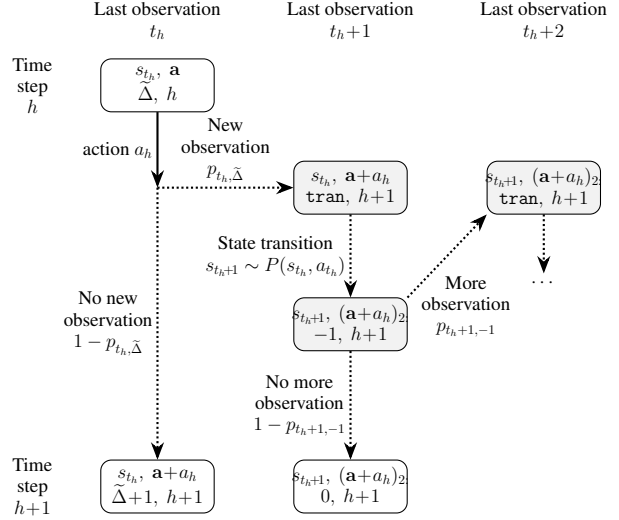


Figure 1. Illustration of the transition of the augmented MDP after taking one action. Gray states indicates intermediate states that have no actions. Straight lines represents the agent’s action and dotted lines represents augmented state transitions. $p_{t_h, \tilde{\Delta}}$ is shorthand for $P_{\text{tran}}(s_{t_h}, a_{t_h}, \tilde{\Delta})$ and $p_{t_h+1, -1}$ is shorthand for $P_{\text{tran}}(s_{t_h+1}, a_{t_h+1}, -1)$, where they show the probability of the transition.

but we introduce two types of intermediate states that allow us to better explain the algorithm and the analysis.

We define the augmented state space as $\mathcal{S}_{\text{aug}} := \mathcal{S} \times \cup_{D=0}^{D_{\max}} \mathcal{A}^D \times \mathcal{D} \times [H+1]$, where $\mathcal{D} := [-1 : \Delta_{\max}] \cup \{\text{tran}\}$. We note that $\mathcal{S}_{\text{aug}}$ is of size exponential in $D_{\max}$, and that any computational implications of this blow-up are apparently unavoidable (see Section 7 for a discussion). Consider an augmented state $s_{\text{aug}} = (s_{t_h}, \mathbf{a}, \tilde{\Delta}_h, h) \in \mathcal{S}_{\text{aug}}$. This state corresponds to the case where the last observed state is s_{t_h} , the unresolved action queue is $\mathbf{a}$, the last action in the queue is a_{h-1} (or possibly empty if $t_h = h$), and $\tilde{\Delta}_h$ time steps have elapsed since s_{t_h} was revealed, but the cases $\tilde{\Delta}_h = -1$ and $\tilde{\Delta}_h = \text{tran}$ convey different meanings. If the initial state of the SDMDP is s_1 , then the initial state of the augmented MDP is $(s_1, \emptyset, 0, 1)$. Depending on $\tilde{\Delta}$, the properties and the transitions of the state are determined. We explain them using three categories.

Category 1: $\tilde{\Delta}_h \in [0 : \Delta_{\max}]$. These augmented states are standard ones where $\tilde{\Delta}_h$ corresponds to the number of time steps elapsed since s_{t_h} was revealed. The augmented-state transition at this state determines whether there will be a new observation or not. When the agent takes an action, the next augmented state is either one of the two: $(s_{t_h}, \mathbf{a}+a, \text{tran}, h+1)$, which corresponds to the case where the next state will be revealed, and $(s_{t_h}, \mathbf{a}+a, \tilde{\Delta}_h+1, h+1)$, which corresponds to not having a new observation. We note that even if a new observation is determined to be revealed, it is not yet revealed at this stage. We define

$P_{\text{tran}} : \mathcal{S} \times \mathcal{A} \times [-1 : D_{\max}] \rightarrow [0, 1]$ as $P_{\text{tran}}(s, a, \tilde{\Delta}) := P_{\Delta \sim P_{\text{delay}}(s, a)}(\Delta = \tilde{\Delta} \mid \Delta \geq \tilde{\Delta}) = \frac{P_{\text{delay}, s, a}(\tilde{\Delta})}{\sum_{\Delta \geq \tilde{\Delta}} P_{\text{delay}, s, a}(\Delta)}$, which represents the probability of transitioning to the first case. Although P_{tran} requires additional information of whether the delay is clipped or the episode has terminated, we omit it here for simplicity.

Category 2: $\tilde{\Delta}_h = \text{tran}$. An augmented state with $\tilde{\Delta}_h = \text{tran}$ is an intermediate state that has no actions, but only state transition. The agent enters this state when the next state is determined to be revealed, and the transition of this augmented state samples the next state. Denoting $a_{t_h} = \mathbf{a}_1$, the next augmented state is $(s', \mathbf{a}_2, -1, h)$ with $s' \sim P_{s_{t_h}, a_{t_h}}$. A reward of $r(s_{t_h}, a_{t_h})$ is received together with this transition.

Category 3: $\tilde{\Delta}_h = -1$. An augmented state with $\tilde{\Delta}_h = -1$ is another type of intermediate state. The agent enters this state immediately after s_{t_h} is revealed, and whether there will be further observations within the same time step is determined at this augmented state. The next augmented state is either $(s_{t_h}, \mathbf{a}, \text{tran}, h)$ with probability $P_{\text{tran}}(s_{t_h}, a_{t_h}, -1)$, which corresponds to the case where another state will be revealed, or $(s_{t_h}, \mathbf{a}, 0, h)$ with the remaining probability.

4.2. Algorithm

After constructing the augmented MDP, we apply standard techniques for common MDPs with an extension using certain structure of the augmented MDP. For the tabular case, we choose MVP (Zhang et al., 2021) as a base algorithm, which is a standard UCBVI-based algorithm with Bernstein-type bonuses. See Algorithm 2 for its optimistic estimation rule. We note that any other UCBVI-based algorithm also works.

As the augmented state space has exponential size in $D_{\max}$, directly applying MVP would lead to exponential regret. The augmented MDP can be learned much faster by noting that many of the transition probabilities are shared based on P or P_{delay} . For example, an observed transition of $(s, \mathbf{a}, \text{tran}, h)$ to $(s', \mathbf{a}_2, -1, h)$ gives information about $P_{s, a}(s')$ that can be used to estimate the transition of $(s, \mathbf{a}', \text{tran}, h)$ for different $\mathbf{a}'$ with $\mathbf{a}'_1 = \mathbf{a}_1$. The core idea is to estimate P and P_{delay} instead of P_{aug} . Hence, instead of storing visit counts of augmented state-action pairs, it is sufficient to store the visit counts of the original state-action pairs as $N^k(s, a) = \sum_{i=1}^{k-1} \sum_{h=1}^H \mathbb{1}\{(s_h^i, a_h^i) = (s, a)\}$ and store an estimate $\hat{P}_{s, a}^k$ of $P_{s, a}$. In the case where P_{delay} is unknown, we additionally store $N^k(s, a, \tilde{\Delta}) = \sum_{i=1}^{k-1} \sum_{h=1}^H \mathbb{1}\{(s_h^i, a_h^i) = (s, a), \Delta_h^i \geq \tilde{\Delta}\}$, which is the number of times that any augmented state of the form $(s, a + \mathbf{a}, \tilde{\Delta}, h)$ is visited. Note that this number is different from the number of $(s, a, \tilde{\Delta})$ in the dataset. We also

estimate P_{tran} by $\hat{P}_{\text{tran}}^k$, defined as

$$\hat{P}_{\text{tran}}^k(s, a, \tilde{\Delta}) := \frac{N^k(s, a, \tilde{\Delta}) - N^k(s, a, \tilde{\Delta} + 1)}{N^k(s, a, \tilde{\Delta})},$$

where the numerator is the number of times the $\tilde{\Delta}$ is sampled from $P_{\text{delay}}(s, a)$, and the denominator is the number of samples from $P_{\text{delay}}(s, a)$ that are greater than or equal to $\tilde{\Delta}$. Using these estimates, we run MVP under a correct update order. Another difference from the standard MVP is that the log factor is increased to a roughly $D \wedge b$ factor, where D is the length of the current action queue, and b is an upper bound on the branching factor of the current augmented state-action pair. An informal description of the algorithm is shown in Algorithm 1. We present the full algorithm that includes the update order and exception handling in Appendix A.

5. Theoretical Guarantees

In this section, we provide regret upper bound results for Algorithm 1. We define $\iota := \log \frac{HSAK}{\delta}$ for logarithmic factors.

Theorem 1. Suppose the delay distribution P_{delay} is known. With probability at least $1 - \delta$, Algorithm 1 achieves the regret bound of $\mathcal{O}(H\sqrt{(D_{\max} \wedge B)SAK}\iota + HBSA\iota^2)$.

Theorem 2. Suppose the delay distribution P_{delay} is unknown. With probability at least $1 - \delta$, Algorithm 1 achieves the regret bound of

$$\mathcal{O}(H\sqrt{(D_{\max} \wedge B)SAK}\iota + H\sqrt{\Delta_{\max}SAK}\iota + H(B + \Delta_{\max})SA\iota^2).$$

Discussion of Theorems 1 and 2. In both known and unknown delay distribution cases, we derive the regret upper bound of $\tilde{\mathcal{O}}(H\sqrt{D_{\max}SAK})$. Compared to the previous bound of $\tilde{\mathcal{O}}(H^{3/2}D_{\max}^{5/2}\sqrt{SAK})$ by Chen et al. (2023), we improve the bound by a factor of $H^{1/2}D_{\max}^2$, significantly improving the $D_{\max}$ -dependency. While the $H^{1/2}$ factor improvement comes from using variance-dependent bonus terms, the $D_{\max}^2$ factor improvement comes from our novel analysis.

Theorems 1 and 2 demonstrate that $D_{\max}$ -dependency can be replaced by a known branching factor B when it is smaller. This fact implies that the performance degradation caused by lengthening the delay is not indefinite. This is because the $D_{\max}$ -dependency arises from taking the union bound over exponentially many states in $D_{\max}$, but it can be replaced by taking the union bound over all bounded functions with domain size B .

In the unknown P_{delay} case, we note that the second term may be larger than the first term only when $B < \Delta_{\max} \leq$

Algorithm 1 MVP-Delayed (Informal)

- 1: Construct augmented MDP $\mathcal{M}_{\text{aug}}$ as in Section 4.1
- 2: Define $\ell^*(D, b) := (D \log A + \log \frac{64H(D+1)^2 \Delta_{\max} SAK}{\delta}) \wedge (b \log \frac{32Hb\Delta_{\max} SAK}{\delta})$
- 3: **for** $k = 1, 2, \dots, K$ **do**
- 4: Compute value estimates as follows for all $s \in \mathcal{S}, \mathbf{a} \in \cup_{D=0}^{D_{\max}} \mathcal{A}^D, \tilde{\Delta} \in [-1 : \Delta_{\max}], h = H, \dots, 1$:
- 5: $V^k(s, \mathbf{a}, \text{tran}, h) \leftarrow \text{MVP-Est} \left(r(s, \mathbf{a}_1), \hat{P}_{s, \mathbf{a}_1}, V^k(\cdot, \mathbf{a}_2, -1, h), N^k(s, \mathbf{a}_1), \ell^*(\text{len}(\mathbf{a}), B) \right)$
- 6: $Q^k((s, \mathbf{a}, \tilde{\Delta}, h), a) \leftarrow \begin{cases} (P_{\text{tran}})_{s, \mathbf{a}', \tilde{\Delta}} V_{\text{tran}} + (1 - (P_{\text{tran}})_{s, \mathbf{a}', \tilde{\Delta}}) V_{\text{delay}} & (\text{known } P_{\text{delay}}) \\ \text{MVP-Est}(0, (\hat{P}_{\text{tran}}^k)_{s, \mathbf{a}', \tilde{\Delta}}, (V_{\text{tran}}, V_{\text{delay}}), N^k(s, \mathbf{a}', \tilde{\Delta}), \ell^*(\text{len}(\mathbf{a}), 2)) & (\text{unknown } P_{\text{delay}}) \end{cases}$
- 7: where $\mathbf{a}' = \mathbf{a} + a, (V_{\text{tran}}, V_{\text{delay}}) = (V^k(s, \mathbf{a}', \text{tran}, h+1), V^k(s, \mathbf{a}', \tilde{\Delta}+1, h+1))$
- 8: $V^k(s, \mathbf{a}, \tilde{\Delta}, h) \leftarrow \max_{a \in \mathcal{A}} Q^k((s, \mathbf{a}, \tilde{\Delta}, h), a)$
- 9: $\pi_h^k(s, \mathbf{a}, \tilde{\Delta}, h) \leftarrow \arg\max_{a \in \mathcal{A}} Q^k((s, \mathbf{a}, \tilde{\Delta}, h), a)$
- 10: Execute π_h^k and collect data $\{s_h^k, a_h^k, \Delta_h^k\}_h$
- 11: **end for**

Algorithm 2 MVP-Est

- 1: **Input:** $r \in \mathbb{R}, \hat{P} \in \Delta(\mathcal{S}), V \in \mathbb{R}^{\mathcal{S}}, N \in \mathbb{N}, \ell \in \mathbb{R}$
- 2: Set constants $c_1 \leftarrow \frac{20}{3}, c_2 \leftarrow \frac{400}{9}$
- 3: **if** $N \leq 1$ **then**
- 4: **Return** H
- 5: **else**
- 6: **Return** $\left(r + \hat{P}V + c_1 \sqrt{\frac{\mathbb{V}(\hat{P}, V)\ell}{N}} + c_2 \frac{H\ell}{N} \right) \wedge H$
- 7: **end if**

$D_{\max}$ holds since $\Delta_{\max} \leq D_{\max}$. In this case, we incur a slightly larger regret as there are $\Delta_{\max} SA$ many values of $P_{\text{tran}}(s, a, \Delta)$ to learn.

Remark 2. The $D_{\max}$ dependency in Theorems 1 and 2 can be improved when the actual lengths of the delays are shorter than $D_{\max}$. Defining $D_{\max}(s, a)$ as the maximum possible length of delay D_h starting from the state-action pair $(s_h, a_h) = (s, a)$, the $(D_{\max} \wedge B)SA$ factor can be reduced to $\sum_{(s,a) \in \mathcal{S} \times \mathcal{A}} (D_{\max}(s, a) \wedge B)$, even if the agent is not aware of individual $D_{\max}(s, a)$ values. In the same way, the $\Delta_{\max} SA$ factor in Theorem 2 may be reduced to $\sum_{(s,a)} \Delta_{\max}(s, a)$, where $\Delta_{\max}(s, a)$ is the unknown maximum inter-arrival time of (s, a) . As a result, even if the agent only knows crude upper bounds of $D_{\max}$ and $\Delta_{\max}$, the regret bound remains the same.

6. Regret Lower Bound

In this section, we provide a regret lower bound result for SDMDPs, showing that our approach is minimax optimal up to logarithmic factors.

Theorem 3 (Lower bound result). *Let $\tilde{D} := \min\{D_{\max}, \frac{H}{4}, \frac{B}{2}, \frac{S}{4} - 1\} = \mathcal{O}(D_{\max} \wedge B)$. Suppose $\tilde{D} \geq c$ for some absolute constant c , $H \geq 8 + 4 \log_A S$, and $K \geq \frac{1}{24} \tilde{D} SA \log \tilde{D}$. Then, for any algorithm, there*

exists a CDMDP instance with S states, A actions, time horizon H , and delay length $D_{\max}$ with branching factor at most B such that the expected regret of the algorithm for K episodes is at least $\Omega(H \sqrt{\tilde{D} SAK} (\log \tilde{D})^{-1})$.

As CDMDPs (constant delay) are special cases of SDMDPs (stochastic delay), the theorem applies to SDMDPs as well.

A lower bound of $\Omega(H \sqrt{SAK})$ can be immediately derived using the hard instance and the analysis for the standard MDP setting by Domingues et al. (2021), as they naturally apply to SDMDPs. From this bound, we show that an additional factor of $D_{\max}^{1/2}$ is necessary in the delayed setting, implying that as the delay becomes longer, the problem becomes statistically harder. Together with Theorem 1, these results establish a tight minimax regret bound up to logarithmic factors, characterizing that the optimal dependence on delay is $D_{\max}^{1/2}$. Theorem 2 also meets this lower bound when $\Delta_{\max} \leq B$. We note that the regret lower bound of $\Omega(\sqrt{HD_{\max} SAK})$ by Chen et al. (2023)² is worse than the standard bound since $D_{\max} \leq H$.

For the proof of Theorem 3, we design a novel structure named *CodeMDP*, which is a CDMDP with delay length D whose learning complexity increases with D . The agent is supposed to find a correct sequence of D actions that maximizes the reward. Using the structure of the CodeMDP, we reduce an ℓ_1 -norm estimation problem of a d -dimensional vector to solving a CodeMDP with unknown transitions. We then establish the following lower bound for estimating the ℓ_1 -norm of a vector, which may be of independent interest.

Proposition 1. *Let $\theta \in [-1, 1]^d$. At each time step t , suppose an index $I_t \sim \text{Unif}([d])$ and a Bernoulli random vari-*

²Proposition 4.3 in Chen et al. (2023) states a regret lower bound of $\Omega(H \sqrt{D_{\max} SAK})$ for the time-inhomogeneous setting, which is translated to $\Omega(\sqrt{HD_{\max} SAK})$ for the time-homogeneous setting we consider.

able $X_t \sim B(\frac{1+\theta_{I_t}}{2})$ are revealed. Then, the sample complexity for estimating $\frac{1}{d}\|\theta\|_1$ up to some additive error ε is at least $\Omega\left(\frac{d}{\varepsilon^2 \log d}\right)$.

An important property of Proposition 1 is that the sample complexity scales with d . As a result, the learning complexity of a CodeMDP also scales with D , which plays a crucial role in obtaining the $D_{\max}^{1/2}$ -factor in Theorem 3. The detailed proof of Theorem 3, including the structure of the Code MDP and its relationship with the problem setting of Proposition 1, is discussed in Appendix D.

7. Computational Hardness

One obstacle in deploying Algorithm 1 is its exponential time complexity in $D_{\max}$. In this section, we provide a negative result regarding the computational hardness of solving delayed MDPs that implies that a polynomial time algorithm is unlikely to exist.

An unobservable MDP (UMDP) is an MDP that the agent cannot observe intermediate states and has to plan H actions in advance. It coincides with CDMDPs with delay $D = H$. More generally, CDMDPs with delay D can be considered as embedding UMDPs with a time horizon D . Assuming that the transition probabilities are given, one method of computing the optimal value of a given UMDP is to compute the values of all possible A^H sequences of actions. There are several results regarding the computational hardness of UMDPs showing that the exponential time complexity is unlikely to be avoidable, and we introduce a result in Burago et al. (1996).

Theorem 4 (Restatement of Theorem 6 in Burago et al. (1996)). *Suppose a UMDP $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, r, H)$ with $H = |\mathcal{S}| = n$ is given. The problem of distinguishing whether the optimal value of the given UMDP is 1 or less than $\exp(-\sqrt{n})$ is NP-hard.*

Theorem 4 states that approximating the optimal value of a UMDP is NP-hard, even if its transition probabilities are fully known. The theorem is proved by reducing a 3-SAT problem, making the problem of approximating the optimal value of SDMDPs strictly harder than 3-SAT problems. Therefore, the exponential time complexity of Algorithm 1 is the best we can hope for.

8. Proof Sketch: Generalization to MDPs with Partially Known Dynamics

In this section, we provide a sketch of how the regret bounds in Section 5 are obtained. Instead of directly proving the results, we first define a more general model which we call MDPs with partially known dynamics. This model captures and generalizes the core properties of the augmented

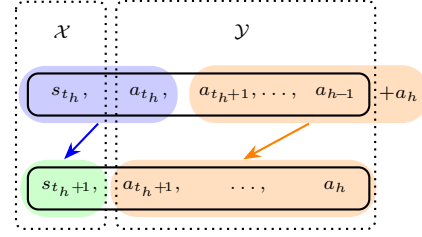


Figure 2. Illustration of core properties of the augmented MDP's state transition. Consider a transition accompanied by an augmented state-action pair of $((s_{t_h}; a_{t_h}, \dots, a_{h-1}), a_h)$. The orange-shaded part indicates that the transition dynamics for the action queue are known, which is simply shifting from the previous action queue. The blue-shaded part indicates that the unknown part of the state transition is determined only by (s_{t_h}, a_{t_h}) and is irrelevant of the other part of the augmented state-action pair.

MDPs we utilize. We provide an algorithm and theoretical guarantees for this setting, and then the results for solving SDMDPs follow as specific cases of the theorem.

8.1. MDPs with Partially Known Dynamics

We first explain the motivation of defining MDPs with partially known dynamics. There are two properties of the augmented MDP that lead to a polynomial regret bound despite its exponential state space. For simpler exposition, assume the constant delay setting, and suppose the augmented state consists only of the last observed state s_{t_h} and the action queue $\mathbf{a} = (a_{t_h}, \dots, a_{h-1})$. The first property of the augmented MDP is that the agent has perfect knowledge about the transition of the action queue — the first action is popped and the current action is pushed — and hence there is nothing to learn. The unknown part of the augmented state transition is the distribution of the next state s_{t_h+1} . The second property of the augmented MDP is that the unknown part of the transition is determined by a relatively small part of the augmented state-action pair. Specifically, the distribution of the next state s_{t_h+1} depends only on (s_{t_h}, a_{t_h}) , and the other part of the augmented state-action pair has no effect on it. In other words, the function class for the unknown part of the transition has a certain structure.

Motivated by these properties, we define an abstract model that has these properties by definition, which may be of independent interest. We formalize the condition we impose on the MDP as follows.

Definition 1 (MDPs with partially known dynamics). *We call an MDP $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, r, H)$ an MDP with partially known dynamics if the following conditions are satisfied. The state space decomposes as $\mathcal{S} = \mathcal{X} \times \mathcal{Y}$. The marginal transition kernel for $\mathcal{Y}$ is known, that is, there exists a known function $P_{\mathcal{Y}} : \mathcal{S} \times \mathcal{A} \rightarrow \Delta(\mathcal{Y})$ such that $\sum_{x \in \mathcal{X}} P_{s,a}(x, y) = (P_{\mathcal{Y}})_{s,a}(y)$. We denote the conditional transition kernel of $\mathcal{X}$ by $P_{\mathcal{X}} : \mathcal{S} \times \mathcal{A} \times \mathcal{Y} \rightarrow \Delta(\mathcal{X})$, which*

satisfies $P_{s,a}(x, y) = (P_{\mathcal{X}})_{s,a,y}(x)(P_{\mathcal{Y}})_{s,a}(y)$. The conditional transition kernel $P_{\mathcal{X}}$ is unknown to the agent but lies in a known function class $\mathcal{P}_{\mathcal{X}} \subset \{P : \mathcal{S} \times \mathcal{A} \times \mathcal{Y} \rightarrow \Delta(\mathcal{X})\}$.

We consider the case where the functions in the function class $\mathcal{P}_{\mathcal{X}}$ depend only on a finite number of features.

Assumption 2 (Tabular structure). $\mathcal{P}_{\mathcal{X}}$ in Definition 1 satisfies the following. There exist a finite set $\mathcal{Z}_{\text{eff}}$ and a known feature map $\phi_{\mathcal{Z}} : \mathcal{S} \times \mathcal{A} \times \mathcal{Y} \rightarrow \mathcal{Z}_{\text{eff}}$ such that $P_{\mathcal{X}} \in \mathcal{P}_{\mathcal{X}}$ only if there exists a distribution $P_{\text{eff}} : \mathcal{Z}_{\text{eff}} \rightarrow \Delta(\mathcal{X})$ with $P_{\mathcal{X}}(s, a, y) = P_{\text{eff}}(\phi_{\mathcal{Z}}(s, a, y))$.

Remark 3. It is important to note that Assumption 2 is not equivalent to MDPs with $\mathcal{Z}_{\text{eff}}$ as a feature space. $\mathcal{Z}_{\text{eff}}$ only represents partial information about the state that governs the unknown part of the state transition. It is possible that multiple state-action pairs are mapped to the same element of $\mathcal{Z}_{\text{eff}}$, but their transition dynamics for the second argument $\mathcal{Y}$ may differ, possibly resulting in drastically different action values for those state-action pairs.

8.2. Algorithm and Theoretical Guarantees

We propose a general algorithm that can run under Assumption 2, using MVP as a base algorithm. Let $B(z) \leq |\mathcal{X}|$ be a known upper bound on the support size of $P_{\text{eff}}(z)$. Algorithm 3 shows the specific procedure. Algorithm 3 has two main differences from learning standard tabular MDPs. First, instead of directly learning the transition distribution P , the algorithm learns P_{eff} , storing the visit count of $z \in \mathcal{Z}_{\text{eff}}$ instead of the actual state-action pairs. Secondly, the logarithmic term is increased by $\log |\mathcal{Y}| \wedge B(z) \log B(z)$, which results from the union bound.

We present the following theorem for Algorithm 3.

Theorem 5. Under Assumption 2, Algorithm 3 achieves the regret bound of

$$\tilde{O}\left(H \sqrt{K \sum_{z \in \mathcal{Z}_{\text{eff}}} ((\log |\mathcal{Y}|) \wedge B(z))} + H \sum_{z \in \mathcal{Z}_{\text{eff}}} B(z)\right).$$

When $B(z) = B$ is a constant, the leading term of Theorem 5 becomes $\tilde{O}(H \sqrt{|\mathcal{Z}_{\text{eff}}| K ((\log |\mathcal{Y}|) \wedge B)})$. Compared to the $\tilde{O}(H \sqrt{SAK})$ regret bound of standard MDPs, the cost of learning SA many transition distributions is reduced to $|\mathcal{Z}_{\text{eff}}|$, but the increased logarithmic factor is reflected in the $(\log |\mathcal{Y}|) \wedge B$ factor. The proof of Theorem 5 is largely based on the standard optimism-based analysis adapted to this setting. The complete proof is presented in Appendix B.

8.3. Application to SDMDPs

In this section, we explain that Theorems 1 and 2 are corollaries of Theorem 5, where rigorous proofs are presented

in Appendix B.4. We mainly focus on the unknown P_{delay} case (Theorem 2) in this section.

We first explain how the augmented MDP constructed in Section 4.1 satisfies Assumption 2. Recall that the augmented state space for a given SDMDP is constructed as $\mathcal{S}_{\text{aug}} = \mathcal{S} \times \cup_{D=0}^{D_{\text{max}}} \mathcal{A}^D \times \mathcal{D} \times [H+1]$. We take $\mathcal{X} = \mathcal{S} \times \mathcal{D}$ and $\mathcal{Y} = \cup_{D=0}^{D_{\text{max}}} \mathcal{A}^D \times [H+1]$. The space $\mathcal{Y}$ consists of the action queue and the time step, and the agent is fully aware of their dynamics. We take $\mathcal{Z}_{\text{eff}} = \mathcal{S} \times \mathcal{A} \times \mathcal{D}$ with $\phi_{\mathcal{Z}}((s, \mathbf{a}, \tilde{\Delta}, h), a, y) = (s, (\mathbf{a} + a)_1, \tilde{\Delta})$. Strictly speaking, we need additional information in the elements of $\mathcal{Z}_{\text{eff}}$ to handle exceptional cases such as the termination of an episode or the truncation of the delay. However, as it only increases the size of $\mathcal{Z}_{\text{eff}}$ by a constant factor, we omit this detail for brevity. P_{eff} is defined as $P_{\text{eff}}(s, a, \text{tran}) = P(s, a) \times \mathbb{1}_{\tilde{\Delta}=-1}$ and $P_{\text{eff}}(s, a, \tilde{\Delta}) = P_{\text{tran}}(s, a, \tilde{\Delta}) \mathbb{1}_{(s, \text{tran})} + (1 - P_{\text{tran}}(s, a, \tilde{\Delta})) \mathbb{1}_{(s, \tilde{\Delta}+1)}$ for $\tilde{\Delta} \in [-1 : \Delta_{\text{max}}]$, where $\mathbb{1}_x$ denotes the point mass distribution at x . In this way, the augmented MDP satisfies Assumption 2.

Now, we explain how Theorem 2 is derived from Theorem 5. There are SA elements of the form $z = (s, a, \text{tran})$. Each of them adds $\log |\mathcal{Y}| \wedge B(z) \lesssim D_{\text{max}} \wedge B$ to the regret bound inside the square root. Therefore, they contribute to the regret bound by $\tilde{O}(H \sqrt{(D_{\text{max}} \wedge B) SAK} + HBSA)$. There are $\mathcal{O}(SA\Delta_{\text{max}})$ remaining elements in $\mathcal{Z}_{\text{eff}}$ of the form $z = (s, a, \tilde{\Delta})$ with $\tilde{\Delta} \in [-1 : \Delta_{\text{max}}]$. We note that they only have a branching factor of two, as they transition to either (s, tran) or $(s, \tilde{\Delta} + 1)$ in $\mathcal{X}$. Consequently, these elements contribute to the $\tilde{O}(H \sqrt{\Delta_{\text{max}} SAK} + H\Delta_{\text{max}} SA)$ terms. Taking the sum over the two bounds yields the bound of Theorem 2.

Theorem 1 is proved by decomposing the state space as $\mathcal{X} = \mathcal{S}$ and $\mathcal{Y} = \cup_{D=0}^{D_{\text{max}}} \mathcal{A}^D \times \mathcal{D} \times [H+1]$, where $\mathcal{D}$ moved from $\mathcal{X}$ to $\mathcal{Y}$ to reflect that the delay distribution is known, and then following the first half of the argument for Theorem 2.

9. Conclusion

We study an efficient method of learning MDPs with delayed state observation. Under the tabular setting, we propose an augmentation-based algorithm and provide a regret bound of $\tilde{O}(H \sqrt{D_{\text{max}} SAK})$, together with a matching regret lower bound, showing that our result is optimal up to logarithmic factors. Our techniques for constructing the augmented MDP and decomposing the known and unknown parts of the transition naturally apply beyond the tabular case, and we believe they provide a general framework for addressing observational delays in RL.

Algorithm 3 Optimistic Algorithm for Partially Known Dynamics

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1: Define  $\ell^*(z) := \log \frac{32H|\mathcal{Y}||\mathcal{Z}_{\text{eff}}|K}{\delta} \wedge B(z) \log \frac{32HB(z)|\mathcal{Z}_{\text{eff}}|K}{\delta}$ 
2: for  $k = 1, 2, \dots$  do
3:    $V_{H+1}^k(s) \leftarrow 0$  for all  $s \in \mathcal{S}$ 
4:   for  $h = H, H-1, \dots, 1$  do
5:     for all  $s \in \mathcal{S}$  do
6:        $Q_h^k(s, a) \leftarrow \mathbb{E}_{\substack{y \sim (P_{\mathcal{Y}})_{(s,a)}, \\ z \leftarrow \phi_{\mathcal{Z}}(s, a, y)}} [\text{MVP-Est}(r(s, a), (\hat{P}_{\text{eff}}^k)_z, V_{h+1}^k(\cdot, y), N^k(z), \ell^*(z))] \text{ for all } a \in \mathcal{A}$ 
7:        $V_h^k(s) \leftarrow \max_{a \in \mathcal{A}} Q_h^k(s, a)$ 
8:        $\pi_h^k(s) \leftarrow \operatorname{argmax}_{a \in \mathcal{A}} Q_h^k(s, a)$ 
9:     end for
10:   end for
11:   Execute  $\pi^k$  and collect data  $\{s_h^k = (x_h^k, y_h^k), a_h^k\}_h$ 
12:   Store  $z_h^k \leftarrow \phi_{\mathcal{Z}}(s_h^k, a_h^k, y_{h+1}^k)$  for  $h \in [H]$ , update  $N^{k+1}(z)$  and  $(\hat{P}_{\text{eff}}^{k+1})_z$  accordingly
13: end for

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Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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A. Full Algorithms

In this section, we present the full procedure of Algorithm 1 introduced in Section 4.2. Algorithm 4 is the full version of Algorithm 1 that specifies the update order and exception handling. As explained in Section 4.2, instead of storing the visit counts of the augmented state-action pairs, it stores the visit counts of the original MDP's state-action pairs and the tuple of state, action, and inter-arrival time. Specifically, it maintains the information of

$$\begin{aligned}
 N^k(s, a) &:= \sum_{i=1}^{k-1} \sum_{h=1}^H \mathbb{1}\{(s_h^i, a_h^i) = (s, a)\} \\
 N^k(s, a, \tilde{\Delta}) &:= \sum_{i=1}^{k-1} \sum_{h=1}^H \mathbb{1}\{(s_h^i, a_h^i) = (s, a), \Delta_h^i \geq \tilde{\Delta}\} \\
 (\hat{P}^k)_{s,a}(s') &:= \frac{1}{N^k(s, a)} \sum_{i=1}^{k-1} \sum_{h=1}^H \mathbb{1}\{(s_h^i, a_h^i, s_{h+1}^i) = (s, a, s')\} \\
 (\hat{P}_{\text{tran}}^k)_{s,a}(\tilde{\Delta}) &:= \frac{N^k(s, a, \tilde{\Delta}) - N^k(s, a, \tilde{\Delta} + 1)}{N^k(s, a, \tilde{\Delta})}.
 \end{aligned} \tag{1}$$

As a subroutine of Algorithm 4, we have Algorithm 5 (Q-Estimate) that handles the computation of Q -values for augmented states with $\tilde{\Delta} \in [-1 : \Delta_{\max}]$. Recall that these augmented states transition to one of the two augmented states depending on whether a new state is revealed or not. When P_{delay} is known, Q-Estimate takes the expectation of the next augmented states' values. When P_{delay} is unknown, MVP-Est is called to compute the UCB value instead. It also handles the case where the next state is trivially revealed due to clipping or the termination of the episode.

Algorithm 4 MVP-Delayed

```

1: Input: Per time step delay  $\Delta_{\max}$ , Maximum delay  $D_{\max}$ 
2: Define  $\ell^*(D, b) := (D \log A + \log \frac{64H(D+1)^2 \Delta_{\max} SAK}{\delta}) \wedge (b \log \frac{32Hb \Delta_{\max} SAK}{\delta})$ 
3: for  $k = 1, 2, \dots$  do
4:   Initialize  $N^k(s, a), N^k(s, a, \tilde{\Delta}), \hat{P}^k(s, a), \hat{P}_{\text{tran}}^k(s, a, \tilde{\Delta})$  as Eq. (1)
5:   Initialize  $V^k(s, \emptyset, \text{tran}, H+1) \leftarrow 0, V^k(s, \emptyset, -1, H+1) \leftarrow 0$  for all  $s \in \mathcal{S}$ 
6:   for all  $s \in \mathcal{S}, \mathbf{a} \in \cup_{D=1}^{D_{\max}} \mathcal{A}^D$  do
7:      $V^k(s, \mathbf{a}, \text{tran}, H+1) \leftarrow \text{MVP-Est}(r(s, \mathbf{a}_1), \hat{P}_{s, \mathbf{a}_1}^k, V^k(\cdot, \mathbf{a}_2, -1, H+1), N^k(s, \mathbf{a}_1), \ell^*(D, B))$ 
8:      $V^k(s, \mathbf{a}, -1, H+1) \leftarrow V^k(s, \mathbf{a}, \text{tran}, H+1)$ 
9:   end for
10:  for  $h = H, \dots, 1$  do
11:    // Update Category 1 states ( $\tilde{\Delta} \in [0 : \Delta_{\max}]$ )
12:    for all  $s \in \mathcal{S}, \mathbf{a} \in \cup_{D=0}^{D_{\max}} \mathcal{A}^D, \tilde{\Delta} \in [0 : \Delta_{\max}]$  do
13:       $Q^k((s, \mathbf{a}, \tilde{\Delta}, h), a) \leftarrow \text{Q-Estimate}(s, \mathbf{a} + a, \tilde{\Delta}, h+1)$  for all  $a \in \mathcal{A}$ 
14:       $V^k(s, \mathbf{a}, \tilde{\Delta}, h) \leftarrow \max_{a \in \mathcal{A}} Q^k((s, \mathbf{a}, \tilde{\Delta}, h), a)$ 
15:       $\pi_h^k(s, \mathbf{a}, \tilde{\Delta}, h) \leftarrow \arg\max_{a \in \mathcal{A}} Q^k((s, \mathbf{a}, \tilde{\Delta}, h), a)$ 
16:    end for
17:    // Update Category 2 and 3 states ( $\tilde{\Delta} = \text{tran or } -1$ )
18:     $V^k(s, \emptyset, -1, h) \leftarrow V^k(s, \emptyset, 0, h)$  for all  $s \in \mathcal{S}$ 
19:    for  $D = 1, \dots, D_{\max}$  do
20:      for all  $s \in \mathcal{S}, \mathbf{a} \in \mathcal{A}^D$  do
21:         $V^k(s, \mathbf{a}, \text{tran}, h) \leftarrow \text{MVP-Est}(r(s, \mathbf{a}_1), \hat{P}_{s, \mathbf{a}_1}^k, V^k(\cdot, \mathbf{a}_2, -1, h), N^k(s, \mathbf{a}_1), \ell^*(D, B))$ 
22:         $V^k(s, \mathbf{a}, -1, h) \leftarrow \text{Q-Estimate}(s, \mathbf{a}, -1, h)$ 
23:      end for
24:    end for
25:  end for
26:  Execute  $\pi^k$  and collect data  $\{s_h^k, a_h^k, \Delta_h^k\}_{h=1}^H \cup \{s_{H+1}^k\}$ 
27: end for

```

Algorithm 5 Q-Estimate

```

1: Input:  $(s, \mathbf{a}, \tilde{\Delta}, h)$ 
2:  $V_{\text{tran}} \leftarrow V^k(s, \mathbf{a}, \text{tran}, h)$ 
3:  $V_{\text{delay}} \leftarrow V^k(s, \mathbf{a}, \tilde{\Delta} + 1, h)$ 
4: if  $\text{len}(\mathbf{a}) = D_{\max} + 1$  or  $\tilde{\Delta} = \Delta_{\max}$  or  $h = H + 1$  then
5:   Return  $V_{\text{tran}}$ 
6: else if  $P_{\text{delay}}$  is known then
7:   Get  $p \leftarrow P_{\text{tran}}(s, \mathbf{a}_1, \tilde{\Delta})$ 
8:   Return  $pV_{\text{tran}} + (1 - p)V_{\text{delay}}$ 
9: else
10:  Return MVP-Est( $0, \hat{P}_{\text{tran}}^k(s, \mathbf{a}_1, \tilde{\Delta}), (V_{\text{tran}}, V_{\text{delay}}), N^k(s, \mathbf{a}_1, \tilde{\Delta}), \ell^*(\text{len}(\mathbf{a}), 2)$ )
11: end if
    
```

B. Proof of Theorem 5

In this section, we present the full proof of Theorem 5.

We define several notations for the analysis. Define $\ell_1 = \log \frac{32H^2|\mathcal{Y}||\mathcal{Z}_{\text{eff}}|K}{\delta}$, $\ell_2(z) = \log \frac{32HB(z)|\mathcal{Z}_{\text{eff}}|K}{\delta}$, and $\ell^*(z) = \min\{\ell_1, B(z)\ell_2(z)\}$. For $z \in \mathcal{Z}_{\text{eff}}$, let $\mathcal{X}(z) := \{x \in \mathcal{X} \mid (P_{\text{eff}})_z(x) > 0\}$ be the support of $(P_{\text{eff}})_z$. Let $N_h^k(z)$ be the number of time steps $z \in \mathcal{Z}_{\text{eff}}$ appeared in the trajectory up to the h -th time step of the k -th episode, that is, $N_h^k(z) := N^k(z) + \sum_{j=1}^h \mathbb{1}\{z_j^k = z\}$. For $h \in [H + 1]$, $s \in \mathcal{S}$, and $k \in [K]$, we define $U_h^k(s)$ and $\tilde{U}_h^k(s)$ iteratively starting from $U_{H+1}^k(s) = \tilde{U}_{H+1}^k(s) := 0$ and

$$\begin{aligned}
 U_h^k(s) &:= \mathbb{E}_{\substack{s'=(x', y') \sim P_{s,a} \\ z=\phi_Z(s, a, y')}} \left[\frac{\ell^*(z)}{N^k(z)} + U_{h+1}^k(s') \right] \wedge 1, \\
 \tilde{U}_h^k(s) &:= \mathbb{E}_{\substack{s'=(x', y') \sim P_{s,a} \\ z=\phi_Z(s, a, y')}} \left[\frac{B(z)\ell_2(z)}{N^k(z)} + \tilde{U}_{h+1}^k(s') \right] \wedge 1,
 \end{aligned}$$

for $h \in [H]$, where $a = \pi_h^k(s)$.

B.1. High-probability Events

In this section, we define the high-probability events that constitute the event $\mathcal{E}$ under which Theorem 5 holds.

Lemma 1. *The following inequality holds for all $z \in \mathcal{Z}_{\text{eff}}$, $y \in \mathcal{Y}$, $h \in [H]$, and $k \in [K]$ that satisfies $N^k(z) \geq 1$ with probability at least $1 - \frac{\delta}{8}$:*

$$\left| ((P_{\text{eff}})_z - (\hat{P}_{\text{eff}})_z) V_{h+1}^*(\cdot, y) \right| \leq \sqrt{\frac{2\mathbb{V}((P_{\text{eff}})_z, V_{h+1}^*(\cdot, y))\ell_1}{N^k(z)}} + \frac{H\ell_1}{3N^k(z)}.$$

Proof. Apply Lemma 25 to $Z = V_{h+1}^*(x, y)$ with $x \sim P_{\text{eff}}(z)$ and the probability of failure as $\frac{\delta}{8}$, then take the union bound over $y \in \mathcal{Y}$, $z \in \mathcal{Z}_{\text{eff}}$, $h \in [H]$, and $1 \leq N^k(s, a) \leq KH$. $\square$

The following lemma extends to arbitrary functions at the cost of $B(z)$ dependence in the bound. The proof is deferred to Appendix C.1.

Lemma 2. *There exists an event with probability at least $1 - \frac{\delta}{8}$ such that for all $z \in \mathcal{Z}_{\text{eff}}$ and for any constants $a \leq b$, the following inequality holds for any function $V : \mathcal{X}(z) \rightarrow [a, b]$:*

$$\left| ((P_{\text{eff}})_z - (\hat{P}_{\text{eff}})_z) V \right| \leq \sqrt{\frac{2\mathbb{V}((P_{\text{eff}})_z, V)B(z)\ell_2(z)}{N^k(z)}} + \frac{(b - a)B(z)\ell_2(z)}{3N^k(z)}.$$

The following Lemmas 3 and 4 are empirical-variance versions of Lemmas 1 and 2, respectively. The proofs of these lemmas are nearly identical to their counterparts, only that the uses of Lemma 25 are replaced by Lemma 26.

Lemma 3. *The following inequality holds for all $z \in \mathcal{Z}_{\text{eff}}$, $y \in \mathcal{Y}$, $h \in [H]$, and $k \in [K]$ that satisfies $N^k(s, a) \geq 1$ with probability at least $1 - \frac{\delta}{8}$:*

$$\left| ((P_{\text{eff}})_z - (\hat{P}_{\text{eff}})_z) V_{h+1}^*(\cdot, y) \right| \leq 2 \sqrt{\frac{\mathbb{V}((\hat{P}_{\text{eff}})_z, V_{h+1}^*(\cdot, y)) \ell_1}{N^k(z)}} + \frac{14H\ell_1}{3N^k(z)}.$$

Lemma 4. *There exists an event with probability at least $1 - \frac{\delta}{8}$ such that for all $z \in \mathcal{Z}_{\text{eff}}$ and for any constants $a \leq b$, the following inequality holds for any function $V : \mathcal{X}(z) \rightarrow [a, b]$:*

$$\left| ((P_{\text{eff}})_z - (\hat{P}_{\text{eff}})_z) V \right| \leq 2 \sqrt{\frac{\mathbb{V}((\hat{P}_{\text{eff}})_z, V) B(z) \ell_2(z)}{N^k(z)}} + \frac{14(b-a)B(z)\ell_2(z)}{3N^k(z)}.$$

We additionally require two lemmas regarding the concentration of $U_1^k(s_1^k)$ and $\tilde{U}_1^k(s_1^k)$ between the sum of $\frac{1}{N^k(z)}$ over the actual trajectory. The following lemma is an application of Lemma 15 in Lee & Oh (2025), and the proof is identical to it.

Lemma 5. *With probability at least $1 - \frac{\delta}{2}$, we have*

$$\sum_{k=1}^K U_1^k \leq 2 \sum_{k=1}^K \sum_{h=1}^H \mathbb{1}\{2N^k(z_h^k) > N_h^k(z_h^k)\} \cdot \frac{\ell^*(z_h^k)}{N^k(z_h^k)} + 3|\mathcal{Z}_{\text{eff}}| \log \frac{8H}{\delta}$$

and

$$\sum_{k=1}^K \tilde{U}_1^k \leq 2 \sum_{k=1}^K \sum_{h=1}^H \mathbb{1}\{2N^k(z_h^k) > N_h^k(z_h^k)\} \cdot \frac{B(z_h^k)\ell_2(z_h^k)}{N^k(z_h^k)} + 3|\mathcal{Z}_{\text{eff}}| \log \frac{8H}{\delta}.$$

We let $\mathcal{E}$ be the intersection of the events of Lemmas 1-5. By the union bound, we have $\mathbb{P}(\mathcal{E}) \geq 1 - \delta$.

B.2. Optimism

In this section, we prove the following optimism lemma for Algorithm 3.

Lemma 6 (Optimism). *The estimated value $V_h^k(s)$ in Algorithm 3 satisfies $V_h^k(s) \geq V_h^*(s)$ for all $s \in \mathcal{S}$ under $\mathcal{E}$.*

To prove the lemma, we require the following lemmas from Zhang et al. (2021).

Lemma 7 (Lemma 14 in Zhang et al. (2021)). *Let $f : \Delta([S]) \times \mathbb{R}^S \times \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ with $f(p, V, n, \ell) := pV + \max\{\frac{20}{3} \sqrt{\frac{\mathbb{V}(p, V)\ell}{n}}, \frac{400}{9} \frac{\ell}{n}\}$, where $\mathbb{V}(p, V) = \sum_{s \in [S]} p(s)(V(s) - pV)^2$. Then, f satisfies*

1. $f(p, V, n, \ell)$ is non-decreasing in $V(s)$ for all $p \in \Delta([S])$, $\|V\|_\infty \leq 1$, $n, \ell > 0$.
2. $f(p, V, n, \ell) \geq pV + 2\sqrt{\frac{\mathbb{V}(p, V)\ell}{n}} + \frac{14\ell}{3n}$.

The following lemma guarantees that MVP-Est outputs an optimistic estimate in Line 8 of Algorithm 3. It follows the same reasoning as Lemma 4 in Zhang et al. (2021).

Lemma 8. *Fix $s \in \mathcal{S}$, $a \in \mathcal{A}$, $y \in \mathcal{Y}$, $h \in [H]$, and $k \in [K]$, and let $z = \phi_{\mathcal{Z}}(s, a, y)$. Define the conditional optimal action value (conditioned on y) as $Q_h^*(s, a, y) = r(s, a) + \sum_{x \in \mathcal{X}} (P_{\mathcal{X}})_{s, a, y}(x) V_{h+1}^*(x, y)$ and the conditional optimistic estimate as $Q_h^k(s, a, y) := \text{MVP-Est}(r(s, a), (\hat{P}_{\text{eff}})_z, V_{h+1}^k(\cdot, y), N^k(z), \ell^*(z))$. Suppose $V_{h+1}^k(x, y) \geq V_{h+1}^*(x, y)$ holds for all $x \in \mathcal{X}$. Then, we have $Q_h^k(s, a, y) \geq Q_h^*(s, a, y)$ under the event $\mathcal{E}$.*

Proof. For simplicity, denote $P(\cdot) := (P_{\text{eff}})_z(\cdot)$, $\hat{P}(\cdot) := (\hat{P}_{\text{eff}})_z(\cdot)$, $r := r(s, a)$, $V_{h+1}^*(\cdot) := V_{h+1}^*(\cdot, y)$, $V_{h+1}^k(\cdot) := V_{h+1}^k(\cdot, y)$, $N := N^k(z)$, and $\ell^* := \ell^*(z)$. If $Q_h^k(s, a, y) = H$, then the lemma becomes trivial. Suppose $Q_h^k(s, a, y) < H$,

which implies $N \geq 2$. Under the event of Lemma 3, we have

$$(P - \hat{P})V_{h+1}^* \leq 2\sqrt{\frac{\mathbb{V}(\hat{P}, V_{h+1}^*)\ell_1}{N}} + \frac{14H\ell_1}{3N}.$$

Under the event of Lemma 4, we have

$$(P - \hat{P})V_{h+1}^* \leq 2\sqrt{\frac{\mathbb{V}(\hat{P}, V_{h+1}^*)B(z)\ell_2(z)}{N}} + \frac{14HB(z)\ell_2(z)}{3N}.$$

Taking the minimum over the two bounds, we have

$$(P - \hat{P})V_{h+1}^* \leq 2\sqrt{\frac{\mathbb{V}(\hat{P}, V_{h+1}^*)\ell^*}{N}} + \frac{14H\ell^*}{3N},$$

where we use that $\ell^* = \ell^*(z) = \ell_1 \wedge B(z)\ell_2(z)$. Then, $Q_h^*(s, a, y)$ is upper bounded as

$$\begin{aligned} Q_h^*(s, a, y) &= r + PV_{h+1}^* \\ &= r + \hat{P}V_{h+1}^* + (P - \hat{P})V_{h+1}^* \\ &\leq r + \hat{P}V_{h+1}^* + 2\sqrt{\frac{\mathbb{V}(\hat{P}, V_{h+1}^*)\ell^*}{N}} + \frac{14H\ell^*}{3N}. \end{aligned}$$

By Lemma 7, we have

$$\begin{aligned} \hat{P}V_{h+1}^* + 2\sqrt{\frac{\mathbb{V}(\hat{P}, V_{h+1}^*)\ell^*}{N}} + \frac{14H\ell^*}{3N} &\leq \hat{P}V_{h+1}^* + \max \left\{ \frac{20}{3} \sqrt{\frac{\mathbb{V}(\hat{P}, V_{h+1}^*)\ell^*}{N}}, \frac{400H\ell^*}{9N} \right\} \\ &\leq \hat{P}V_{h+1}^k + \max \left\{ \frac{20}{3} \sqrt{\frac{\mathbb{V}(\hat{P}, V_{h+1}^k)\ell^*}{N}}, \frac{400H\ell^*}{9N} \right\} \\ &\leq \hat{P}V_{h+1}^k + \frac{20}{3} \sqrt{\frac{\mathbb{V}(\hat{P}, V_{h+1}^k)\ell^*}{N}} + \frac{400H\ell^*}{9N}, \end{aligned}$$

where we use the condition $V_{h+1}^k \geq V_{h+1}^*$ for the second inequality. Therefore, we conclude that

$$\begin{aligned} Q_h^*(s, a, y) &\leq r + \hat{P}V_{h+1}^* + 2\sqrt{\frac{\mathbb{V}(\hat{P}, V_{h+1}^*)\ell^*}{N}} + \frac{14H\ell^*}{3N} \\ &\leq r + \hat{P}V_{h+1}^k + \frac{20}{3} \sqrt{\frac{\mathbb{V}(\hat{P}, V_{h+1}^k)\ell^*}{N}} + \frac{400H\ell^*}{9N} \\ &= Q_h^k(s, a, y). \end{aligned}$$

□

We now prove Lemma 6.

Proof of Lemma 6. We prove the lemma by backward induction on h . The inequality is trivial for $h = H + 1$. Suppose the inequality holds for $h + 1$. Then, by Lemma 8, we have $Q_h^k(s, a, y) \geq Q_h^*(s, a, y)$. It implies that

$$\begin{aligned} Q_h^k(s, a) &= \sum_{y \in \mathcal{Y}} P_{\mathcal{Y}}(y | s, a) Q_h^k(s, a, y) \\ &\geq \sum_{y \in \mathcal{Y}} P_{\mathcal{Y}}(y | s, a) Q_h^*(s, a, y) \\ &= Q_h^*(s, a). \end{aligned}$$

Since $V_h^k(s)$ is the maximum of $Q_h^k(s, a)$ over $a \in \mathcal{A}$, we have

$$V_h^k(s) = \max_{a \in \mathcal{A}} Q_h^k(s, a) \geq \max_{a \in \mathcal{A}} Q_h^*(s, a) = V_h^*(s).$$

This completes the induction step. $\square$

B.3. Bounding Cumulative Regret

For the remainder of the analysis, we combine the techniques developed in [Zhang et al. \(2024\)](#); [Lee & Oh \(2025\)](#) for simpler analysis. We complete the proof of Theorem 5 by proving the following lemma:

Lemma 9. *Under $\mathcal{E}$, the following inequality holds for all $k \in [K]$:*

$$\sum_{k=1}^K \left(V_1^k(s_1^k) - V_1^{\pi^k}(s_1^k) \right) \leq 56H \sqrt{K(\log eKH) \sum_{z \in \mathcal{Z}_{\text{eff}}} \ell^*(z)} + 2640H(\log eKH) \sum_{z \in \mathcal{Z}_{\text{eff}}} B(z)\ell_2(z).$$

Proof of Lemma 9. Recall that U_h^k and $\tilde{U}_h^k$ are functions defined as

$$U_h^k(s) := \mathbb{E}_{\substack{s'=(x', y') \sim P_{s,a} \\ z=\phi_Z(s, a, y')}} \left[\frac{\ell^*(z)}{N^k(z)} + U_{h+1}^k(s') \right] \wedge 1,$$

$$\tilde{U}_h^k(s) := \mathbb{E}_{\substack{s'=(x', y') \sim P_{s,a} \\ z=\phi_Z(s, a, y')}} \left[\frac{B(z)\ell_2(z)}{N^k(z)} + \tilde{U}_{h+1}^k(s') \right] \wedge 1,$$

for $h \in [H]$, where $a = \pi_h^k(s)$, starting from $U_{H+1}^k(s) = \tilde{U}_{H+1}^k(s) := 0$. The following lemma shows that the instantaneous regret is upper bounded by U_1^k and $\tilde{U}_1^k$.

Lemma 10. *Under $\mathcal{E}$, we have*

$$V_1^k(s_1^k) - V_1^{\pi^k}(s_1^k) \leq 28H \sqrt{U_1^k(s_1^k)} + 660H \tilde{U}_1^k(s_1^k)$$

for all $k \in [K]$.

We provide a high level proof sketch of Lemma 10 here. We express the instantaneous regret as the sum of the bonus terms and some extra terms as

$$V_1^k(s_1^k) - V_1^{\pi^k}(s_1^k) \lesssim \mathbb{E}_{\pi^k} \left[\sum_{h=1}^H \left(\sqrt{\frac{\mathbb{V}((P_{\text{eff}})_{z_h^k}, V_{h+1}^*(\cdot, y_{h+1}^k) \ell^*(z_h^k))}{N^k(z_h^k)}} + \frac{HB(z_h^k)\ell_2(z_h^k)}{N^k(z_h^k)} \right) \right].$$

By applying the Cauchy-Schwarz inequality twice on the first term, we obtain that

$$\mathbb{E}_{\pi^k} \left[\sum_{h=1}^H \sqrt{\frac{\mathbb{V}((P_{\text{eff}})_{z_h^k}, V_{h+1}^*(\cdot, y_{h+1}^k) \ell^*(z_h^k))}{N^k(z_h^k)}} \right] \leq \sqrt{\mathbb{E}_{\pi^k} \left[\sum_{h=1}^H \mathbb{V}((P_{\text{eff}})_{z_h^k}, V_{h+1}^*(\cdot, y_{h+1}^k)) \right]} \mathbb{E}_{\pi^k} \left[\sum_{h=1}^H \frac{\ell^*(z_h^k)}{N^k(z_h^k)} \right].$$

Then, we show that the expected sum of the variances are upper bounded by H^2 , and the sum of the $\frac{\ell^*(z)}{N^k(z)}$ -type terms are expressed as U_1^k and $\tilde{U}_1^k$, which yields Lemma 10. The full proof of Lemma 10 is deferred to Appendix C.2.

Taking the sum over $k \in [K]$ and applying Lemma 10, we obtain that

$$\begin{aligned} \sum_{k=1}^K \left(V_1^k(s_1^k) - V_1^{\pi^k}(s_1^k) \right) &\leq \sum_{k=1}^K \left(28H \sqrt{U_1^k(s_1^k)} + 660H \tilde{U}_1^k(s_1^k) \right) \\ &\leq 28H \sqrt{K \sum_{k=1}^K U_1^k(s_1^k)} + 660H \sum_{k=1}^K \tilde{U}_1^k(s_1^k), \end{aligned}$$

where we use the Cauchy-Schwarz inequality for the second inequality. The sums of $U_1^k(s_1^k)$ and $\tilde{U}_1^k(s_1^k)$, which are the expected sums of $\frac{1}{N(z)}$ -type of terms, are bounded by the following lemma, whose proof is deferred to Appendix C.3.

Lemma 11. *Under $\mathcal{E}$, we have*

$$\sum_{k=1}^K U_1^k(s_1^k) \leq 4(\log eKH) \sum_{z \in \mathcal{Z}_{\text{eff}}} \ell^*(z)$$

and

$$\sum_{k=1}^K \tilde{U}_1^k(s_1^k) \leq 4(\log eKH) \sum_{z \in \mathcal{Z}_{\text{eff}}} B(z) \ell_2(z).$$

By applying Lemma 11, we have

$$\begin{aligned} \sum_{k=1}^K \left(V_1^k(s_1^k) - V_1^{\pi^k}(s_1^k) \right) &\leq 28H \sqrt{K \sum_{k=1}^K U_1^k(s_1^k) + 660H \sum_{k=1}^K \tilde{U}_1^k(s_1^k)} \\ &\leq 56H \sqrt{K(\log eKH) \sum_{z \in \mathcal{Z}_{\text{eff}}} \ell^*(z) + 2640H(\log eKH) \sum_{z \in \mathcal{Z}_{\text{eff}}} B(z) \ell_2(z)}, \end{aligned}$$

completing the proof. $\square$

Proof of Theorem 5. By Lemma 6 and Lemma 9, we have

$$\begin{aligned} \sum_{k=1}^K \left(V_1^*(s_1^k) - V_1^{\pi^k}(s_1^k) \right) &\leq \sum_{k=1}^K \left(V_1^k(s_1^k) - V_1^{\pi^k}(s_1^k) \right) \\ &= \mathcal{O} \left(H \sqrt{K(\log KH) \sum_{z \in \mathcal{Z}_{\text{eff}}} \ell^*(z)} + H(\log KH) \sum_{z \in \mathcal{Z}_{\text{eff}}} B(z) \ell_2(z) \right). \end{aligned} \quad (2)$$

Recall that $\ell^*(z) = \log \frac{32H^2|\mathcal{Y}||\mathcal{Z}_{\text{eff}}|K}{\delta} \wedge B(z) \log \frac{32HB(z)|\mathcal{Z}_{\text{eff}}|K}{\delta}$ and $\ell_2(z) = \log \frac{32HB(z)|\mathcal{Z}_{\text{eff}}|K}{\delta}$. Omitting logarithmic factors on $H, K, \delta, B(z)$, and $|\mathcal{Z}_{\text{eff}}|$, we conclude that the regret bound is in the order of

$$\tilde{\mathcal{O}} \left(H \sqrt{K \sum_{z \in \mathcal{Z}_{\text{eff}}} (\log |\mathcal{Y}|) \wedge B(z)} + H \sum_{z \in \mathcal{Z}_{\text{eff}}} B(z) \right).$$

$\square$

B.4. Proof of Theorems 1 and 2

Proof of Theorem 1. We first show that the augmented MDP constructed in Section 4.1 satisfies Assumption 2. Recall that the augmented state space is defined as $\mathcal{S}_{\text{aug}} = \mathcal{S} \times \cup_{D=0}^{D_{\text{max}}} \mathcal{A} \times \mathcal{D} \times [H+1]$. We decompose it as $\mathcal{X} = \mathcal{S}$ and $\mathcal{Y} = \cup_{D=0}^{D_{\text{max}}} \mathcal{A} \times \mathcal{D} \times [H+1]$. Since the agent knows the dynamics of the action queue, the delay distribution, and the current time step, the agent is fully aware of the dynamics of $\mathcal{Y}$. We take $\mathcal{Z}_{\text{eff}} = \mathcal{S} \times (\mathcal{A} \cup \{\emptyset\})$, where the feature map is defined as $\phi_{\mathcal{Z}}((s, \mathbf{a}, \text{tran}, h), a, y) = (s, \mathbf{a}_1)$ and $\phi_{\mathcal{Z}}((s, \mathbf{a}, \tilde{\Delta}, h), a, y) = (s, \emptyset)$ for $\tilde{\Delta} \neq \text{tran}$. The effective transition distribution becomes $P_{\text{eff}}(s, a) = P(s, a)$ for $a \in \mathcal{A}$ and $P_{\text{eff}}(s, \emptyset) = \mathbb{1}_s$. In this way, we have shown that the augmented MDP satisfies Assumption 2.

Now, we show that Theorem 1 is a corollary of Theorem 5. We have $|\mathcal{Y}| \leq (A+1)^{D_{\text{max}}} (\Delta_{\text{max}} + 2)(H+1)$, and hence $\log |\mathcal{Y}| \lesssim D_{\text{max}}$, omitting the logarithmic factors. For $z \in \mathcal{Z}_{\text{eff}}$, we have $B(z) \leq B$. We also have $|\mathcal{Z}_{\text{eff}}| \leq S(A+1)$.

Applying Theorem 5, where we may refer to Eq. (2) for logarithmic dependence, we conclude that the regret bound is at most

$$\mathcal{O}\left(H\sqrt{((D_{\max}\log A + \iota) \wedge (B\iota))SAK(\log KH)} + HBSA(\log KH)\iota\right),$$

where $\iota = \log \frac{HSAK}{\delta}$.

In order to achieve a tighter problem-dependent bound that scales with the actual length of the delay as mentioned in Remark 2, we take the union bound in a specially designed way. Instead of taking the uniform union bound over $\mathcal{Y}$, we assign probabilities depending on the length of $\mathbf{a}$. Specifically, while we previously took the union bound by assigning the same probability of $\frac{\delta'}{(A+1)^{D_{\max}}}$ to all $\mathbf{a} \in \cup_{D=0}^{D_{\max}} \mathcal{A}^D$, instead, we assign probability $\frac{\delta'}{2(D+1)^2 A^D}$ to action queues $\mathbf{a}$ in $\mathcal{A}^D$. By doing so, the $D_{\max} \log A$ factor reduces to $D \log A$ for such elements. In the analysis, only the reachable states affect the regret bound. Hence, defining $D_{\max}(s, a)$ as the maximum possible delay length of D_h followed by $(s_h, a_h) = (s, a)$ for any $h \in [H]$, we achieve the regret bound whose $(D_{\max} \wedge B)SA$ factor is replaced by $\sum_{(s,a) \in \mathcal{S} \times \mathcal{A}} (D_{\max}(s, a) \wedge B)$. $\square$

Proof of Theorem 2. Section 8.3 largely explains the main processes, and we fill in some missing details. To show that the augmented MDP satisfies Assumption 2, we decompose the state space as $\mathcal{X} = \mathcal{S} \times \mathcal{D}$ and $\mathcal{Y} = \cup_{D=0}^{D_{\max}} \mathcal{A}^D \times [H+1]$. We take $\mathcal{Z}_{\text{eff}} = \mathcal{S} \times \mathcal{A} \times \mathcal{D} \times \{\emptyset, 0, 1\}$, where we need the last component to handle exceptional cases such as truncation of the delay or the termination of the episode. For most augmented states, we have $\phi_{\mathcal{Z}}((s, \mathbf{a}, \tilde{\Delta}, h), a, y) = (s, \mathbf{a}_1, \tilde{\Delta}, \emptyset)$. The last component is set to 0 when we have $\text{len}(\mathbf{a}) = 0$ and $\tilde{\Delta} = -1$, meaning that there is no state to reveal and the delay must be set to 0. The last component is set to 1 when we have either $\text{len}(\mathbf{a}) = D_{\max}$ and $\tilde{\Delta} \in [0 : \Delta_{\max}]$, which corresponds to case that the delay is truncated by $D_{\max}$, or $h = H+1$, which corresponds to the termination of the episode.

The effective transition distribution is defined as $P_{\text{eff}}(s, a, \text{tran}, \cdot) = P_{s,a} \times \mathbb{1}_{\tilde{\Delta}=-1}$ and $P_{\text{eff}}(s, a, \tilde{\Delta}, \emptyset) = P_{\text{tran}}(s, a, \tilde{\Delta})\mathbb{1}_{s,a,\text{tran}} + (1 - P_{\text{tran}}(s, a, \tilde{\Delta}))\mathbb{1}_{s,a,\tilde{\Delta}+1}$ for $\tilde{\Delta} \neq \text{tran}$. If the last variable is 0 or 1 in the latter case, then P_{tran} is replaced by 0 or 1, respectively. In this way, we can correctly classify the augmented MDP as a special case of Assumption 2.

Showing that Theorem 2 follows from Theorem 5 is exactly as explained in Section 8.3, where we obtain the regret bound of

$$\begin{aligned} &\mathcal{O}\left(H\sqrt{((D_{\max}\log A + \iota) \wedge (B\iota) + \Delta_{\max}\iota)SAK(\log KH)}\right. \\ &\quad \left.+ H(B + \Delta_{\max})SA(\log KH)\iota\right). \end{aligned}$$

Achieving the improved problem-dependent bound described in Remark 2 is done in the same way with Theorem 1. $\square$

C. Proofs of Technical Lemmas in Appendix B

C.1. Proof of Lemma 2

We first prove the following lemma.

Lemma 12. *The following inequality holds for all $x \in \mathcal{X}$, $z \in \mathcal{Z}_{\text{eff}}$, and $k \in [K]$ that satisfies $N^k(z) \geq 2$ with probability at least $1 - \frac{\delta}{8}$:*

$$\left| (P_{\text{eff}})_z(x) - (\hat{P}_{\text{eff}}^k)_z(x) \right| \leq \sqrt{\frac{2(P_{\text{eff}})_z(x)\ell_2(z)}{N^k(z)}} + \frac{\ell_2(z)}{3N^k(z)}.$$

Proof. Fix $x \in \mathcal{X}$, $z \in \mathcal{Z}_{\text{eff}}$, and $n \geq 2$. If $(P_{\text{eff}})_z(x) = 0$, then the inequality is trivial. Suppose $x \in \mathcal{X}(z)$. Apply Lemma 25 to $Z = \mathbb{1}\{x' = x\}$ with $x' \sim P_{\text{eff}}(z)$. Note that in this case, we have $\text{Var}(Z) = (P_{\text{eff}})_z(x)(1 - (P_{\text{eff}})_z(x)) \leq (P_{\text{eff}})_z(x)$. Then, we obtain that the following inequality holds with probability at least $1 - \frac{\delta}{8}$:

$$\left| (P_{\text{eff}})_z(x) - (\hat{P}_{\text{eff}}^k)_z(x) \right| \leq \sqrt{\frac{2(P_{\text{eff}})_z(x) \log \frac{32}{\delta}}{N^k(z)}} + \frac{\log \frac{32}{\delta}}{3N^k(z)}.$$

The proof is completed by taking the union bound over $z \in \mathcal{Z}_{\text{eff}}$, corresponding $x \in \mathcal{X}(z)$, and $2 \leq n \leq KH$, where we use that $|\mathcal{X}(z)| \leq B(z)$ by the definition of $B(z)$. $\square$

Proof of Lemma 2. Fix $z \in \mathcal{Z}_{\text{eff}}$ and $k \in [K]$ with $N^k(z) \geq 1$. Since $\sum_{x \in \mathcal{X}(z)} (P_{\text{eff}})_z(x) = \sum_{x \in \mathcal{X}(z)} (\hat{P}_{\text{eff}}^k)_z(x) = 1$, we have

$$\left((P_{\text{eff}})_z - (\hat{P}_{\text{eff}}^k)_z \right) V = \left((P_{\text{eff}})_z - (\hat{P}_{\text{eff}}^k)_z \right) (V - (P_{\text{eff}})_z V),$$

and by the triangle inequality,

$$\left| \left((P_{\text{eff}})_z - (\hat{P}_{\text{eff}}^k)_z \right) (V - (P_{\text{eff}})_z V) \right| \leq \sum_{x \in \mathcal{X}(z)} \left| \left((P_{\text{eff}})_z(x) - (\hat{P}_{\text{eff}}^k)_z(x) \right) (V(x) - (P_{\text{eff}})_z V) \right|.$$

Under the event of Lemma 12, for all $x \in \mathcal{X}$, we have

$$\left| (P_{\text{eff}})_z(x) - (\hat{P}_{\text{eff}}^k)_z(x) \right| \leq \sqrt{\frac{2(P_{\text{eff}})_z(x) \ell_2(z)}{N^k(z)}} + \frac{\ell_2(z)}{3N^k(z)}.$$

Multiplying both sides by $|V(x) - (P_{\text{eff}})_z V|$ and summing over $x \in \mathcal{X}(z)$ gives

$$\begin{aligned} & \sum_{x \in \mathcal{X}(z)} \left| \left((P_{\text{eff}})_z(x) - (\hat{P}_{\text{eff}}^k)_z(x) \right) (V(x) - (P_{\text{eff}})_z V) \right| \\ & \leq \sum_{x \in \mathcal{X}(z)} \left(\sqrt{\frac{2(P_{\text{eff}})_z(x) \ell_2(z)}{N^k(z)}} |V(x) - (P_{\text{eff}})_z V| + \frac{\ell_2(z)}{3N^k(z)} |V(x) - (P_{\text{eff}})_z V| \right) \\ & \leq \sum_{x \in \mathcal{X}(z)} \sqrt{\frac{2(P_{\text{eff}})_z(x) (V(x) - (P_{\text{eff}})_z V)^2 \ell_2(z)}{N^k(z)}} + \frac{\ell_2(z)}{3N^k(z)} \sum_{x \in \mathcal{X}(z)} |V(x) - (P_{\text{eff}})_z V| \\ & \leq \sqrt{\frac{2B(z) \ell_2(z)}{N^k(z)} \sum_{x \in \mathcal{X}(z)} (P_{\text{eff}})_z(x) (V(x) - (P_{\text{eff}})_z V)^2} + \frac{\ell_2(z)}{3N^k(z)} \cdot B(z)(b-a) \\ & = \sqrt{\frac{2\mathbb{V}((P_{\text{eff}})_z, V) B(z) \ell_2(z)}{N^k(z)}} + \frac{(b-a)B(z) \ell_2(z)}{3N^k(z)}, \end{aligned}$$

where we use the Cauchy-Schwarz inequality and that $|V(x) - (P_{\text{eff}})_z V| \leq b-a$ in the third inequality. $\square$

C.2. Proof of Lemma 10

For the proof of Lemma 10, we require the following lemmas.

Lemma 13. Fix $s \in \mathcal{S}$, $h \in [H]$, and $a = \pi_h^k(s)$. Under $\mathcal{E}$, we have

$$\begin{aligned} V_h^k(s) - r(s, a) - P_{s,a} V_{h+1}^k & \leq \frac{1}{2H} (P_{s,a} (V_{h+1}^k - V_{h+1}^*)^2 - (V_h^k(s) - V_h^*(s))^2) \\ & \quad + \mathbb{E}_{\substack{y' \sim P_{\mathcal{Y}}(s,a) \\ z = \phi_{\mathcal{Z}}(s,a,y')}} \left[28 \sqrt{\frac{\mathbb{V}((P_{\text{eff}})_z, V_{h+1}^*(\cdot, y')) \ell^*(z)}{N^k(z)}} + \frac{660HB(z) \ell_2(z)}{N^k(z)} \right]. \end{aligned}$$

The proof of Lemma 13 is quite technical, and it is deferred to Appendix C.4. Using Lemma 13, we can prove the following bound for the per-episode regret.

Lemma 14. Under $\mathcal{E}$, we have

$$V_1^k(s_1^k) - V_1^{\pi^k}(s_1^k) \leq 28 \sqrt{\mathbb{E}_{\pi^k} \left[\sum_{h=1}^H \mathbb{V}((P_{\text{eff}})_{z_h^k}, V_{h+1}^*(\cdot, y_{h+1}^k)) \right]} U_1^k(s_1^k) + 660H \tilde{U}_1^k(s_1^k)$$

for all $k \in [K]$.

Proof. We define

$$W_h^k(s) := \mathbb{E}_{\pi^k(\cdot|s_h=s)} \left[\sum_{j=h}^H \mathbb{V}((P_{\text{eff}})_{z_j^k}, V_{j+1}^*(\cdot, y_{j+1}^k)) \right],$$

which is the expectation of the sum of variances. Note that we have $W_h^k(s) = \mathbb{E}_{s'=(x', y') \sim P_{s,a}} [\mathbb{V}((P_{\text{eff}})_z, V_{h+1}^*(\cdot, y')) + U_{h+1}^k(s')]$. We prove that the following inequality holds for all $h \in [H]$ and $s \in \mathcal{S}$ by backward induction on h :

$$\begin{aligned} V_h^k(s) - V_h^{\pi^k}(s) &\leq 28\sqrt{W_h^k(s)U_h^k(s)} + 660H\tilde{U}_h^k(s) \\ &\quad - \frac{1}{2H} (V_h^k(s) - V_h^*(s))^2. \end{aligned}$$

The inequality is trivial for $h = H + 1$. Suppose the inequality holds for $h + 1$. We note that the inequality is also trivial when $\tilde{U}_h^k(s) = 1$, as the left-hand side is at most H , while the right-hand side becomes larger than H . Suppose $\tilde{U}_h^k(s) < 1$. We note that we have $U_h^k(s) \leq \tilde{U}_h^k$, hence we also have $U_h^k(s) < 1$. We perform the following decomposition:

$$V_h^k(s) - V_h^{\pi^k}(s) = \underbrace{V_h^k(s) - r(s, a) - P_{s,a}V_{h+1}^k}_{I_1} + \underbrace{P_{s,a}(V_{h+1}^k - V_{h+1}^*)}_{I_2}.$$

I_1 is bounded by Lemma 13 as

$$\begin{aligned} I_1 &\leq \underbrace{\frac{1}{2H} (P_{s,a}(V_{h+1}^k - V_{h+1}^*)^2 - (V_h^k(s) - V_h^*(s))^2)}_{I_3} \\ &\quad + \underbrace{\mathbb{E}_{\substack{y' \sim P_{\mathcal{Y}}(s,a) \\ z = \phi_Z(s,a,y')}} \left[28\sqrt{\frac{\mathbb{V}((P_{\text{eff}})_z, V_{h+1}^*(\cdot, y'))\ell^*(z)}{N^k(z)}} \right]}_{I_4} + \underbrace{\mathbb{E}_{\substack{y' \sim P_{\mathcal{Y}}(s,a) \\ z = \phi_Z(s,a,y')}} \left[\frac{660HB(z)\ell_2(z)}{N^k(z)} \right]}_{I_5}. \end{aligned}$$

I_2 is bounded by the induction hypothesis as

$$I_2 \leq \underbrace{\mathbb{E}_{s' \sim P_{s,a}} \left[28\sqrt{W_{h+1}^k(s')U_{h+1}^k(s')} \right]}_{I_6} + \underbrace{660HP_{s,a}\tilde{U}_{h+1}^k}_{I_7} - \underbrace{\frac{1}{2H}P_{s,a}(V_{h+1}^k - V_{h+1}^*)^2}_{I_8}.$$

We have $I_3 - I_8 = -\frac{1}{2H}(V_h^k(s) - V_h^*(s))^2$ and $I_5 + I_7 = 660H\tilde{U}_h^k(s)$. It remains to bound $I_4 + I_6$. Using the Cauchy-Schwarz inequality, we obtain

$$\begin{aligned} &\sqrt{\frac{\mathbb{V}((P_{\text{eff}})_z, V_{h+1}^*(\cdot, y'))\ell^*(z)}{N^k(z)}} + \sqrt{W_{h+1}^k(s')U_{h+1}^k(s')} \\ &\leq \sqrt{\mathbb{V}((P_{\text{eff}})_z, V_{h+1}^*(\cdot, y')) + W_{h+1}^k(s')} \sqrt{\frac{\ell^*(z)}{N^k(z)} + U_{h+1}^k(s')}. \end{aligned}$$

By the Cauchy-Schwarz inequality for expectation and the law of total expectation, we obtain that

$$\begin{aligned}
 I_4 + I_6 &\leq 28 \mathbb{E}_{\substack{s'=(x',y') \sim P_{s,a} \\ z=\phi_Z(s,a,y')}} \left[\sqrt{\mathbb{V}((P_{\text{eff}})_z, V_{h+1}^*(\cdot, y')) + W_{h+1}^k(s')} \sqrt{\frac{\ell^*(z)}{N^k(z)} + U_{h+1}^k(s')} \right] \\
 &\leq 28 \sqrt{\mathbb{E}_{\substack{s'=(x',y') \sim P_{s,a} \\ z=\phi_Z(s,a,y')}} [\mathbb{V}((P_{\text{eff}})_z, V_{h+1}^*(\cdot, y')) + W_{h+1}^k(s')]} \\
 &\quad \times \sqrt{\mathbb{E}_{\substack{s'=(x',y') \sim P_{s,a} \\ z=\phi_Z(s,a,y')}} \left[\frac{\ell^*(z)}{N^k(z)} + U_{h+1}^k(s') \right]} \\
 &= 28 \sqrt{W_h^k(s) U_h^k(s)}.
 \end{aligned}$$

Combining the bounds for $I_3 - I_8$ and $I_5 + I_7$, we obtain the desired bound for $V_h^k(s) - V_h^{\pi^k}(s)$, completing the induction step. $\square$

Proof of Lemma 10. Given Lemma 14, it suffices to prove that $\mathbb{E}_{\pi^k} \left[\sum_{h=1}^H \mathbb{V}((P_{\text{eff}})_{z_h^k}, V_{h+1}^*(\cdot, y_{h+1}^k)) \right] \leq H^2$. By Lemma 17, we have

$$\begin{aligned}
 &\mathbb{E}_{y_{h+1}^k \sim P_{\mathcal{Y}}(s_h^k, a_h^k)} \left[\mathbb{V}((P_{\text{eff}})_{z_h^k}, V_{h+1}^*(\cdot, y_{h+1}^k)) \right] \\
 &\leq P_{s_h^k, a_h^k}(V_{h+1}^*)^2 - (V_h^*(s_h^k))^2 + 2H \max\{V_h^*(s_h^k) - P_{s_h^k, a_h^k} V_{h+1}^*, 0\} \\
 &= P_{s_h^k, a_h^k}(V_{h+1}^*)^2 - (V_h^*(s_h^k))^2 + 2H (V_h^*(s_h^k) - P_{s_h^k, a_h^k} V_{h+1}^*),
 \end{aligned}$$

where we use that $V_h^*(s_h^k) - P_{s_h^k, a_h^k} V_{h+1}^* \geq Q_h^*(s_h^k) - P_{s_h^k, a_h^k} V_{h+1}^* = r(s_h^k, a_h^k) \geq 0$ for the last equality. Taking the sum over $h \in [H]$ and taking the expectation, the terms are telescoped and we derive

$$\begin{aligned}
 \mathbb{E}_{\pi^k} \left[\sum_{h=1}^H \mathbb{V}((P_{\text{eff}})_{z_h^k}, V_{h+1}^*(\cdot, y_{h+1}^k)) \right] &\leq -(V_1^*(s_1^k))^2 + 2H V_1^*(s_1^k) \\
 &= H^2 - (V_1^*(s_1^k) - H)^2 \\
 &\leq H^2.
 \end{aligned}$$

Plugging this bound into Lemma 14, we obtain Lemma 10. $\square$

C.3. Proof of Lemma 11

Proof of Lemma 11. Under the event of Lemma 5, we have

$$\sum_{k=1}^K U_1^k(s_1^k) \leq 2 \sum_{k=1}^K \sum_{h=1}^H \mathbb{1}\{2N^k(z_h^k) > N_h^k(z_h^k)\} \frac{\ell^*(z_h^k)}{N^k(z_h^k)} + 3|\mathcal{Z}_{\text{eff}}| \log \frac{8H}{\delta}.$$

Under $2N^k(z_h^k) > N_h^k(z_h^k)$, we have $\frac{1}{N^k(z_h^k)} \leq \frac{2}{N_h^k(z_h^k)}$, and we also have $N_h^k(z_h^k) \geq 2$, and hence the first sum is bounded as follows:

$$\begin{aligned}
 2 \sum_{k=1}^K \sum_{h=1}^H \mathbb{1}\{2N^k(z_h^k) > N_h^k(z_h^k)\} \frac{\ell^*(z_h^k)}{N^k(z_h^k)} &\leq \sum_{k=1}^K \sum_{h=1}^H \frac{4\mathbb{1}\{N_h^k(z_h^k) \geq 2\} \ell^*(z_h^k)}{N_h^k(z_h^k)} \\
 &= 4 \sum_{z \in \mathcal{Z}_{\text{eff}}} \ell^*(z) \sum_{n=2}^{N^{K+1}(z)} \frac{1}{n} \\
 &\leq 4 \sum_{z \in \mathcal{Z}_{\text{eff}}} \ell^*(z) (\log KH).
 \end{aligned}$$

We add $3|\mathcal{Z}_{\text{eff}}| \log \frac{8H}{\delta}$ and using that $\log \frac{8H}{\delta} \leq \ell^*(z)$, we derive that

$$\begin{aligned} \sum_{k=1}^K U_1^k(s_1^k) &\leq 4 \sum_{z \in \mathcal{Z}_{\text{eff}}} \ell^*(z) (\log KH) + 3|\mathcal{Z}_{\text{eff}}| \log \frac{8H}{\delta} \\ &\leq 4 \sum_{z \in \mathcal{Z}_{\text{eff}}} (\ell^*(z) (\log KH) + 1) \\ &= 4(\log eKH) \sum_{z \in \mathcal{Z}_{\text{eff}}} \ell^*(z), \end{aligned}$$

proving the first inequality. The second inequality is proved in the exact same way. $\square$

C.4. Proof of Lemma 13

We require two additional lemmas to prove Lemma 13.

Lemma 15. *Under $\mathcal{E}$, the following inequality holds for all $z \in \mathcal{Z}_{\text{eff}}$, $y \in \mathcal{Y}$, $h \in [H]$, and $k \in [K]$ with $N^k(z) \geq 2$:*

$$\begin{aligned} &\frac{20}{3} \sqrt{\frac{\mathbb{V}((\hat{P}_{\text{eff}})_z V_{h+1}^k(\cdot, y)) \ell^*(z)}{N^k(z)}} + \frac{400H \ell^*(z)}{9N^k(z)} \\ &\leq 12 \sqrt{\frac{\mathbb{V}((P_{\text{eff}})_z, V_{h+1}^*(\cdot, y)) \ell^*(z)}{N^k(z)}} + \frac{1}{8H} \mathbb{V}((P_{\text{eff}})_z, (V_{h+1}^k - V_{h+1}^*)(\cdot, y)) + \frac{325HB(z)\ell_2(z)}{N^k(z)}. \end{aligned}$$

Proof. We fix $z \in \mathcal{Z}_{\text{eff}}$, $y \in \mathcal{Y}$, $h \in [H]$, and $k \in [K]$. Denote $P := (P_{\text{eff}})_z$, $\hat{P} := (\hat{P}_{\text{eff}})_z$, $V_{h+1}^* := V_{h+1}^*(\cdot, y)$, $V_{h+1}^k := V_{h+1}^k(\cdot, y)$, $N := N^k(z)$, and $\ell^* := \ell^*(z)$ for simplicity. Under the event of Lemma 2, by applying the lemma to $V(\cdot) = (V_{h+1}^k(\cdot) - PV_{h+1}^k)^2$ we have

$$\begin{aligned} (\hat{P} - P)(V_{h+1}^k - PV_{h+1}^k)^2 &\leq \sqrt{\frac{2\mathbb{V}(P, (V_{h+1}^k - PV_{h+1}^k)^2)B(z)\ell_2(z)}{N}} + \frac{H^2B(z)\ell_2(z)}{3N} \\ &\leq \frac{1}{2H^2} \mathbb{V}(P, (V_{h+1}^k - PV_{h+1}^k)^2) + \frac{4H^2B(z)\ell_2(z)}{3N} \\ &\leq \frac{1}{2} P(V_{h+1}^k - PV_{h+1}^k)^2 + \frac{4H^2B(z)\ell_2(z)}{3N}, \end{aligned} \tag{3}$$

where we use the AM-GM inequality for the second inequality and that $\text{Var}(Z) \leq \mathbb{E}[Z^2] \leq c\mathbb{E}[Z]$ for random variables $Z \in [0, c]$ for the third inequality. Then, we bound the empirical variance of V_{h+1}^k as follows:

$$\begin{aligned} \mathbb{V}(\hat{P}, V_{h+1}^k) &= \hat{P} \left(V_{h+1}^k - \hat{P} V_{h+1}^k \right)^2 \\ &\leq \hat{P} \left(V_{h+1}^k - P V_{h+1}^k \right)^2 \\ &\leq \frac{3}{2} P \left(V_{h+1}^k - P V_{h+1}^k \right)^2 + \frac{4H^2B(z)\ell_2(z)}{3N} \\ &= \frac{3}{2} \mathbb{V}(P, V_{h+1}^k) + \frac{4H^2B(z)\ell_2(z)}{3N} \\ &\leq 3\mathbb{V}(P, V_{h+1}^*) + 3\mathbb{V}(P, V_{h+1}^k - V_{h+1}^*) + \frac{4H^2B(z)\ell_2(z)}{3N}, \end{aligned}$$

where the second inequality uses inequality (3), and the last inequality uses that $\text{Var}(X + Y) \leq 2\text{Var}(X) + 2\text{Var}(Y)$ for

any random variables X and Y . Applying this bound, we have

$$\begin{aligned}
 & \sqrt{\frac{\mathbb{V}(\hat{P}, V_{h+1}^k) \ell^*}{N}} \\
 & \leq \sqrt{\frac{\left(3\mathbb{V}(P, V_{h+1}^*) + 3\mathbb{V}(P, V_{h+1}^k - V_{h+1}^*) + \frac{4H^2 B(z) \ell_2(z)}{3N}\right) \ell^*}{N}} \\
 & \leq \sqrt{\frac{3\mathbb{V}(P, V_{h+1}^*) \ell^*}{N}} + \sqrt{\frac{3\mathbb{V}(P, V_{h+1}^k - V_{h+1}^*) \ell^*}{N}} + \frac{2HB(z) \ell_2(z)}{\sqrt{3N}}, \tag{4}
 \end{aligned}$$

where the last inequality uses that $\sqrt{a+b+c} \leq \sqrt{a} + \sqrt{b} + \sqrt{c}$, and $\ell^*(z) \leq B(z) \ell_2(z)$ for the last term. For the second term, we apply the AM-GM inequality and obtain

$$\begin{aligned}
 \sqrt{\frac{3\mathbb{V}(P, V_{h+1}^k - V_{h+1}^*) \ell^*}{N}} & \leq \frac{3\mathbb{V}(P, V_{h+1}^k - V_{h+1}^*)}{160H} + \frac{40H\ell^*}{N} \\
 & \leq \frac{3\mathbb{V}(P, V_{h+1}^k - V_{h+1}^*)}{160H} + \frac{40HB(z) \ell_2(z)}{N}. \tag{5}
 \end{aligned}$$

Combining inequalities (4) and (5), we have

$$\sqrt{\frac{\mathbb{V}(\hat{P}, V_{h+1}^k) \ell^*}{N}} \leq \sqrt{\frac{3\mathbb{V}(P, V_{h+1}^*) \ell^*}{N}} + \frac{3\mathbb{V}(P, V_{h+1}^k - V_{h+1}^*)}{160H} + \frac{42HB(z) \ell_2(z)}{N}.$$

Finally, we obtain that

$$\begin{aligned}
 \frac{20}{3} \sqrt{\frac{\mathbb{V}(\hat{P} V_{h+1}^k) \ell^*}{N}} + \frac{400H\ell^*}{9N} & \leq 12 \sqrt{\frac{\mathbb{V}(P, V_{h+1}^*) \ell^*}{N}} + \frac{1}{8H} \mathbb{V}(P, V_{h+1}^k - V_{h+1}^*) + \frac{280HB(z) \ell_2(z)}{N} + \frac{400H\ell^*}{9N} \\
 & \leq 12 \sqrt{\frac{\mathbb{V}(P, V_{h+1}^*) \ell^*}{N}} + \frac{1}{8H} \mathbb{V}(P, V_{h+1}^k - V_{h+1}^*) + \frac{325HB(z) \ell_2(z)}{N}.
 \end{aligned}$$

□

Lemma 16. Under $\mathcal{E}$, we have that for all $z \in \mathcal{Z}_{\text{eff}}$, $y \in \mathcal{Y}$, $h \in [H]$, and $k \in [K]$ with $N^k(z) \geq 1$, we have

$$((\hat{P}_{\text{eff}}^k)_z - (P_{\text{eff}})_z) V_{h+1}^k(\cdot, y) \leq \sqrt{\frac{2\mathbb{V}((P_{\text{eff}})_z, V_{h+1}^*(\cdot, y)) \ell^*(z)}{N^k(z)}} + \frac{1}{8H} \mathbb{V}((P_{\text{eff}})_z, (V_{h+1}^k - V_{h+1}^*)(\cdot, y)) + \frac{14HB(z) \ell_2(z)}{3N^k(z)}.$$

Proof. Denote $P := (P_{\text{eff}})_z$, $\hat{P} := (\hat{P}_{\text{eff}})_z$, $V_{h+1}^* := V_{h+1}^*(\cdot, y)$, $V_{h+1}^k := V_{h+1}^k(\cdot, y)$, $N := N^k(z)$, and $\ell^* := \ell^*(z)$. We begin by adding and subtracting $(\hat{P} - P)V_{h+1}^*$.

$$(\hat{P} - P)V_{h+1}^k = (\hat{P} - P)V_{h+1}^* + (\hat{P} - P)(V_{h+1}^k - V_{h+1}^*). \tag{6}$$

By Lemmas 1 and 2, we have

$$(\hat{P} - P)V_{h+1}^* \leq \sqrt{\frac{2\mathbb{V}(P, V_{h+1}^*) \ell^*}{N}} + \frac{H\ell^*}{3N}, \tag{7}$$

where we use that $\ell^* := \ell^*(z) = \ell_1 \wedge B(z) \ell_2(z)$. By Lemma 2, and using that $0 \leq V_{h+1}^k(x) - V_{h+1}^*(x) \leq H$ for all $x \in \mathcal{X}$ by Lemma 6, we have

$$\begin{aligned}
 (\hat{P} - P)(V_{h+1}^k - V_{h+1}^*) & \leq \sqrt{\frac{2\mathbb{V}(P, V_{h+1}^k - V_{h+1}^*) B(z) \ell_2(z)}{N}} + \frac{HB(z) \ell_2(z)}{3N} \\
 & \leq \frac{1}{8H} \mathbb{V}(P, V_{h+1}^k - V_{h+1}^*) + \frac{13HB(z) \ell_2(z)}{3N}, \tag{8}
 \end{aligned}$$

where we use the AM-GM inequality for the last inequality. Plugging inequalities (7) and (8) into inequality (6), we obtain that

$$\begin{aligned} (\hat{P} - P)V_{h+1}^k &\leq \sqrt{\frac{2\mathbb{V}(P, V_{h+1}^*)\ell^*}{N}} + \frac{H\ell^*}{3N} + \frac{1}{8H}\mathbb{V}(P, V_{h+1}^k - V_{h+1}^*) + \frac{13HB(z)\ell_2(z)}{3N} \\ &\leq \sqrt{\frac{2\mathbb{V}(P, V_{h+1}^*)\ell^*}{N}} + \frac{1}{8H}\mathbb{V}(P, V_{h+1}^k - V_{h+1}^*) + \frac{14HB(z)\ell_2(z)}{3N}, \end{aligned}$$

where we use that $\ell^* \leq B(z)\ell_2(z)$ by definition for the last inequality. $\square$

Proof of Lemma 13. We let $b_h^k(s, a, y') := \frac{20}{3}\sqrt{\frac{\mathbb{V}((\hat{P}_{\text{eff}})_z, V_{h+1}^k(\cdot, y')\ell^*(z))}{N^k(z)}} + \frac{400H\ell^*(z)}{N^k(z)}$, where $z = \phi_{\mathcal{Z}}(s, a, y')$. The value of $b_h^k(s, a, y')$ corresponds to the optimistic bonus term added by MVP-Est. We first expand $V_h^k(s)$ as follows:

$$\begin{aligned} V_h^k(s) &= \mathbb{E}_{y' \sim P_{\mathcal{Y}}(s, a)}[Q_h^k(s, a, y')] \\ &\leq \mathbb{E}_{y' \sim P_{\mathcal{Y}}(s, a)}\left[r(s, a) + (\hat{P}_{\text{eff}}^k)_z V_{h+1}^k(\cdot, y') + b_h^k(s, a, y')\right] \\ &= r(s, a) + \mathbb{E}_{y' \sim P_{\mathcal{Y}}(s, a)}\left[(\hat{P}_{\text{eff}}^k)_z V_{h+1}^k(\cdot, y') + b_h^k(s, a, y')\right]. \end{aligned}$$

Then, we obtain that

$$V_h^k(s) - r(s, a) - P_{s,a}V_{h+1}^k \leq \mathbb{E}_{y' \sim P_{\mathcal{Y}}(s, a)}\left[b_h^k(s, a, y') + ((\hat{P}_{\text{eff}}^k)_z - (P_{\text{eff}})_z)V_{h+1}^k(\cdot, y')\right].$$

By Lemmas 15 and 16, we have

$$\begin{aligned} &b_h^k(s, a, y') + ((\hat{P}_{\text{eff}}^k)_z - (P_{\text{eff}})_z)V_{h+1}^k(\cdot, y') \\ &\leq 14\sqrt{\frac{\mathbb{V}((P_{\text{eff}})_z, V_{h+1}^*(\cdot, y'))\ell^*(z)}{N^k(z)}} + \frac{1}{4H}\mathbb{V}((P_{\text{eff}})_z, (V_{h+1}^k - V_{h+1}^*)(\cdot, y')) + \frac{330HB(z)\ell_2(z)}{N^k(z)}. \end{aligned}$$

Let $I_1 := \max\{V_h^k(s) - r(s, a) - P_{s,a}V_{h+1}^k, 0\}$. Then, we obtain

$$I_1 \leq \mathbb{E}_{y' \sim P_{\mathcal{Y}}(s, a)}\left[14\sqrt{\frac{\mathbb{V}((P_{\text{eff}})_z, V_{h+1}^*(\cdot, y'))\ell^*(z)}{N^k(z)}} + \frac{1}{4H}\mathbb{V}((P_{\text{eff}})_z, (V_{h+1}^k - V_{h+1}^*)(\cdot, y')) + \frac{330HB(z)\ell_2(z)}{N^k(z)}\right]. \quad (9)$$

Now, we focus on the second term in the expectation. Denoting $\tilde{V}_h^k(s) := V_h^k(s) - V_h^*(s)$, we know that $0 \leq \tilde{V}_h^k(s) \leq H$ by Lemma 6 and bounds on $V_h^k(s) \leq H$ and $V_h^*(s) \geq 0$. By Lemma 17, we have

$$\mathbb{V}((P_{\text{eff}})_z, \tilde{V}_{h+1}^k(\cdot, y')) \leq P_{s,a}(\tilde{V}_{h+1}^k)^2 - (\tilde{V}_h(s))^2 + 2H \max\left\{\tilde{V}_h^k(s) - P_{s,a}\tilde{V}_{h+1}^k, 0\right\}.$$

Note that we have

$$\begin{aligned} \tilde{V}_h^k(s) - P_{s,a}\tilde{V}_{h+1}^k &= V_h^k(s) - V_h^*(s) - P_{s,a}(V_{h+1}^k - V_{h+1}^*) \\ &\leq V_h^k(s) - Q_h^*(s, a) - P_{s,a}V_{h+1}^k + P_{s,a}V_{h+1}^* \\ &= V_h^k(s) - r(s, a) - P_{s,a}V_{h+1}^k, \end{aligned}$$

and hence

$$\mathbb{V}((P_{\text{eff}})_z, \tilde{V}_{h+1}^k(\cdot, y')) \leq P_{s,a}(\tilde{V}_{h+1}^k)^2 - (\tilde{V}_h^k(s))^2 + 2HI_1.$$

Plugging this bound into inequality (9), we obtain

$$I_1 \leq \frac{1}{2}I_1 + \frac{1}{4H} (P_{s,a}(V_{h+1}^k - V_{h+1}^*)^2 - (V_h^k(s) - V_h^*(s))^2) \\ + \mathbb{E}_{y' \sim P_{\mathcal{Y}}(s,a)} \left[14 \sqrt{\frac{\mathbb{V}((P_{\text{eff}})_z, V_{h+1}^*(\cdot, y')) \ell^*}{N^k(z)}} + \frac{330HB(z)\ell_2(z)}{N^k(z)} \right].$$

Solving the inequality with respect to I_1 , we conclude that

$$I_1 \leq \frac{1}{2H} (P_{s,a}(V_{h+1}^k - V_{h+1}^*)^2 - (V_h^k(s) - V_h^*(s))^2) \\ + \mathbb{E}_{y' \sim P_{\mathcal{Y}}(s,a)} \left[28 \sqrt{\frac{\mathbb{V}((P_{\text{eff}})_z, V_{h+1}^*(\cdot, y')) \ell^*}{N^k(z)}} + \frac{660HB(z)\ell_2(z)}{N^k(z)} \right].$$

□

C.5. Additional Technical Lemmas

Lemma 17. *Let $c \geq 0$ be a constant. Let $\{V_h\}_{h=1}^{H+1}$ be a sequence of functions with $V_h : \mathcal{S} \rightarrow [0, c]$. Then, for any $(s, a) \in \mathcal{S} \times \mathcal{A}$ and $h \in [H]$, we have*

$$\mathbb{E}_{y' \sim P_{\mathcal{Y}}(s,a)} \mathbb{V}((P_{\text{eff}})_{\phi_Z(s,a,y')}, V_{h+1}(\cdot, y')) \leq P_{s,a}(V_{h+1})^2 - (V_h(s))^2 + 2c \max\{V_h(s) - P_{s,a}V_{h+1}, 0\}.$$

Proof. By the law of total variance, we have

$$\mathbb{E}_{y' \sim P_{\mathcal{Y}}(s,a)} \mathbb{V}((P_{\text{eff}})_{\phi_Z(s,a,y')}, V_{h+1}(\cdot, y')) = \mathbb{V}(P_{s,a}, V_{h+1}) - \mathbb{V}((P_{\mathcal{Y}})_{s,a}, (P_{\text{eff}})_{\phi_Z(s,a,y')} V_{h+1}(\cdot, y')) \\ \leq \mathbb{V}(P_{s,a}, V_{h+1}).$$

Applying Lemma 18 to $\mathbb{V}(P_{s,a}, V_{h+1})$ completes the proof. □

Lemma 18 (Lemma 27 in Lee & Oh (2025)). *Let $c \geq 0$ be a constant. Let $\{V_h\}_{h=1}^{H+1}$ sequences of functions with $V_h : \mathcal{S} \rightarrow [0, c]$. Then, for any $(s, a) \in \mathcal{S} \times \mathcal{A}$, we have*

$$\mathbb{V}(P_{s,a}, V_{h+1}) \leq P_{s,a}(V_{h+1})^2 - (V_h(s))^2 + 2c \max\{V_h(s) - P_{s,a}V_{h+1}, 0\}.$$

D. Proof of Theorem 3

In this section, we prove the lower bound result.

D.1. Hard Structure for Delayed Observation: CodeMDP

In this section, we describe a specially designed MDP with delayed observation, whose learning complexity increases with the length of the delay. This structure serves as a core component of the hard instances for the regret lower bound.

Given D , a *CodeMDP* is an MDP with a state space $\mathcal{S}_{\text{code}} = ([D] \times \{0, 1\}) \cup \{s_{\text{succ}}, s_{\text{fail}}\}$ and an action space $\mathcal{A} = \{0, 1\}$. The transition of the CodeMDP is deterministic and is as follows:

- For states (i, b) with $i = 2, \dots, D$ and $b = 0, 1$, any action takes the agent to state $(i - 1, b)$.
- For states $(1, b)$ with $b = 0, 1$, taking action b leads to s_{succ} , and taking the other action leads to s_{fail} .
- For state s_{succ} , taking any action yields a reward of 1 and returns to s_{succ} .
- For state s_{fail} , taking any action yields a reward of 0 and returns to s_{fail} .

Due to the delay, the agent cannot observe the intermediate states. The result is only revealed after the episode, or at least after entering either s_{succ} or s_{fail} , so the agent must plan a sequence of actions $a_1, \dots, a_D$ in advance. The mechanism of CodeMDP is as follows. The agent randomly arrives at an initial state, say (i, b) , but agent can not observe the arrived state. The agent takes the planned sequence of actions $a_1, \dots, a_D$. After taking $i - 1$ actions, the agent arrives at $(1, b)$, which is unobservable, and then takes the action a_i . If $a_i = b$, then the agent transitions to s_{succ} and receives the reward, otherwise fails.

The randomness of the CodeMDP that the agent must learn is the initial state distribution. For instance, suppose the initial state is distributed uniformly over states $\{(i, 0)\}_{i=1}^D$, then the agent receives the reward if and only if $a_i = 0$, so the optimal sequence of actions is $(0, \dots, 0)$. In general, suppose the initial state is sampled from a fixed distribution P over $\mathcal{S}_{\text{code}}$, where $P(s_{\text{succ}}) = P(s_{\text{fail}}) = 0$. For the sake of brevity, we momentarily assume that s_{succ} transitions to s_{fail} after receiving a reward, so that the total reward is always 0 or 1. Then, the expected reward, or the Q -value of an action sequence $\mathbf{a} = (a_1, \dots, a_D)$ under P is $Q_P(\mathbf{a}) = \sum_{i=1}^D \sum_{b \in \{0,1\}} P(i, b) \mathbb{1}\{a_i = b\} = \sum_{i=1}^D P(i, a_i)$. It follows that the optimal actions are $a_i^* = \arg\max_{b \in \{0,1\}} P(i, b)$ for $i = 1, \dots, D$, and the optimal value function is

$$V^*(P) := \sum_{i=1}^D \max\{P(i, 0), P(i, 1)\}.$$

In this sense, P encodes a correct ‘‘codeword’’ $(a_1^*, \dots, a_D^*)$ that yields the maximum reward of $V^*(P)$.

Now, we consider a certain parameterization of P . For $\boldsymbol{\theta} \in [-1, 1]^D$, let $P(\boldsymbol{\theta})$ assign probability $\frac{1-\theta_i}{2D}$ to state $(i, 0)$ and $\frac{1+\theta_i}{2D}$ to state $(i, 1)$. The codeword for this distribution is $(\mathbb{1}\{\theta_1 \geq 0\}, \dots, \mathbb{1}\{\theta_D \geq 0\})$ with the optimal value being

$$\begin{aligned} V^*(P(\boldsymbol{\theta})) &= \sum_{i=1}^D \max\left\{\frac{1+\theta_i}{2D}, \frac{1-\theta_i}{2D}\right\} \\ &= \frac{1}{2} + \frac{1}{2D} \sum_{i=1}^D |\theta_i| \\ &= \frac{1}{2} + \frac{1}{2D} \|\boldsymbol{\theta}\|_1. \end{aligned}$$

Suppose there is an additional initial state s_0 with A actions. Suppose there are A vectors $\{\boldsymbol{\theta}_a\}_{a \in \mathcal{A}}$ such that $P_{s_0, a} = P(\boldsymbol{\theta}_a)$. Then, the optimal action from s_0 is the action a^* with the largest $\|\boldsymbol{\theta}_{a^*}\|_1$, and taking any other action incurs instantaneous regret of at least $\frac{1}{2D}(\|\boldsymbol{\theta}_{a^*}\|_1 - \|\boldsymbol{\theta}_a\|_1)$, let alone taking the correct codeword actions of $P(\boldsymbol{\theta}_a)$. Then, the agent must find the optimal arm by finding the vector $\boldsymbol{\theta}_a$ with the largest ℓ_1 -norm. This problem can be reduced to the following problem formulation, which is also the one described in Proposition 1.

ℓ_1 -norm estimation problem. Let $\boldsymbol{\theta} \in [-1, 1]^d$. For $t = 1, \dots, n$, an index $I_t \sim \text{Unif}([d])$ is sampled, and then $X_t \sim \text{Ber}(\frac{1+\theta_{I_t}}{2})$ is sampled. Based on the observations $\{(I_t, X_t)\}_{t=1}^n$, what is the minimum number of n required to estimate $\frac{1}{d}\|\boldsymbol{\theta}\|_1$ up to an additive error of ε ?

The observation (I_t, X_t) in the ℓ_1 -norm estimation problem corresponds to the state $(I_t, X_t) \in \mathcal{S}_{\text{code}}$ being sampled as the first state of the CodeMDP. Although the agent cannot observe that state immediately, it is observed after the end of the episode, which is why we can identify the observation types of the ℓ_1 -norm estimation and the CodeMDP. Note that the ℓ_1 -norm estimation problem only models the learning of $V^*(P(\boldsymbol{\theta}_a))$ and disregards the learning of the optimal action sequence. We later show that the hardness of learning $V^*(P(\boldsymbol{\theta}))$ is enough to prove Theorem 3.

We restate Proposition 1, which states a lower bound for the sample complexity of the ℓ_1 -norm estimation problem.

Proposition 2 (Formal statement of Proposition 1). *Suppose d is larger than some absolute constant and $0 < \varepsilon < \frac{1}{48 \log d}$ is given. Let $n = \lfloor \frac{d}{1152\varepsilon^2 \log d} \rfloor$. Suppose $\hat{L}^n$ is an estimator for $\frac{1}{d}\|\boldsymbol{\theta}\|_1$ based on n observations $\{(I_t, X_t)\}_{t=1}^n$. Then, we have*

$$\inf_{\hat{L}^n} \sup_{\boldsymbol{\theta} \in [-1, 1]^d} \mathbb{P}_{\boldsymbol{\theta}} \left(\left| \hat{L}^n - \frac{1}{d}\|\boldsymbol{\theta}\|_1 \right| \geq \varepsilon \right) \geq \frac{1}{8},$$

meaning that every estimator based on only n samples incurs the error of at least ε for some $\boldsymbol{\theta}$ with probability at least $\frac{1}{8}$.

The proposition is closely related to Theorem 3 in Cai & Low (2011). Their theorem also considers the problem of estimating $\|\theta\|_1$, but the observation considered is a single d -dimensional vector sampled from a Gaussian distribution $\mathcal{N}(\theta, I_d)$, and the measurement of risk is also different. The proof of Proposition 2 follows the high-level roadmap of Theorem 3 in Cai & Low (2011), but the intermediate steps require a significantly different type of computation due to this difference. The proof of Proposition 2 is presented in Appendix E.

D.2. Construction of Hard Instances

Recall the structure where there is an initial state s_0 with A actions that transition to a CodeMDP, whose transition probabilities are parameterized by θ_a . Suppose exactly one of the $\|\theta_a\|_1$ is larger than the others by ε . Proposition 2 implies that the agent must take each action at least $\Omega(\frac{D}{\varepsilon^2 \log D})$ times to specify the optimal action, and failing to do so will incur $\Omega(\varepsilon K)$ regret. Tuning ε leads to a (non-rigorous) lower bound of $\Omega(\sqrt{\frac{DAK}{\log D}})$. We inflate this bound by a $\sqrt{S}$ factor by using the tree structure considered in Domingues et al. (2021). Hence, the final structure of the hard instances will be a combination of the tree structure and the CodeMDP, where the initial state of the whole MDP will be the root of the tree. We explain the structure of the tree in more detail.

The tree part of the MDP is a A -ary tree of states with $\Theta(S)$ leaves. Specifically, we choose a set of leaves $\mathcal{L} \subset \mathcal{S}$ with $|\mathcal{L}|$ equal to the power of 2 that satisfies $\frac{S}{8} \leq |\mathcal{L}| < \frac{S}{4}$. Then, we construct the tree bottom-up, adding parent states, one for at most A children. It can be shown that the height of the tree is at most $H_{\text{tree}} \leq 1 + \lceil \log_A \frac{S}{4} \rceil$ and the total number of states in the tree is at most $\frac{S}{2}$. For a state in the tree that is not a leaf, each action corresponds to one of its children, where the action transitions to that child. Then, the path from the root to a leaf corresponds to a sequence of $H_{\text{tree}} - 1$ actions that takes the agent deterministically from the root to the leaf state.

Then, we add a CodeMDP structure with $D = \tilde{D} := \min\{D_{\max}, \frac{H}{4}, \frac{B}{2}, \frac{S}{4} - 1\}$. For each leaf-action pair $(l, a) \in \mathcal{L}, \mathcal{A}$, we assign a vector $\theta_{l,a} \in [-1, 1]^d$ so that $P_{l,a} = P(\theta_{l,a})$, which is a distribution over $\mathcal{S}_{\text{code}}$ defined in Appendix D.1. One of the goals of the agent becomes to find the pair that has the largest $\frac{1}{D} \|\theta_{l,a}\|_1$ based on the observation. Figure 3 illustrates the structure of the whole MDP.

We assume $H \geq 4H_{\text{tree}} \geq 8 + 4 \log_A S$, so that taking H_{tree} actions to reach a leaf state and then taking $\tilde{D} \leq \frac{H}{4}$ many actions for the CodeMDP takes at most $\frac{H}{2}$ time steps. We note that the total number of states required to construct this structure is at most $\frac{S}{2} + 2\tilde{D} + 2 \leq S$, and the branching factor is $2\tilde{D} \leq B$.

D.3. Proof of Theorem 3

In this section, we prove Theorem 3 using the tools we developed.

We assumed that s_{succ} transitions to s_{fail} in Appendix D.1 for brevity. We revert the assumption and assume that s_{succ} returns to itself. By doing so, the total reward the agent receives per episode scales with H . However, we must additionally address that the total reward may vary depending on *when* the agent enters the success state. For instance, if the initial state of the CodeMDP is $(1, b)$, then the agent enters s_{succ} after one action, whereas starting from the state $(\tilde{D}, b)$ takes $\tilde{D}$ time steps to enter s_{succ} . Let $\tilde{H} = H - H_{\text{tree}}$. Then, if the agent enters the state (i, b) in the CodeMDP through one of the leaf states, the agent receives the reward of $\tilde{H} - i$, assuming it successfully enters s_{succ} . Then, the optimal value function of a given distribution P over $\mathcal{S}_{\text{code}}$ changes to

$$V^*(P) := \sum_{i=1}^{\tilde{D}} (\tilde{H} - i) \max\{P(i, 0), P(i, 1)\}.$$

Considering $P(\theta)$ parameterized by $\theta \in [-1, 1]^{\tilde{D}}$, we have

$$\begin{aligned} V^*(P(\theta)) &= \sum_{i=1}^{\tilde{D}} (\tilde{H} - i) \frac{1 + |\theta_i|}{2\tilde{D}} \\ &= \frac{1}{2} \sum_{i=1}^{\tilde{D}} (\tilde{H} - i) + \frac{1}{2\tilde{D}} \sum_{i=1}^{\tilde{D}} (\tilde{H} - i) |\theta_i|. \end{aligned}$$

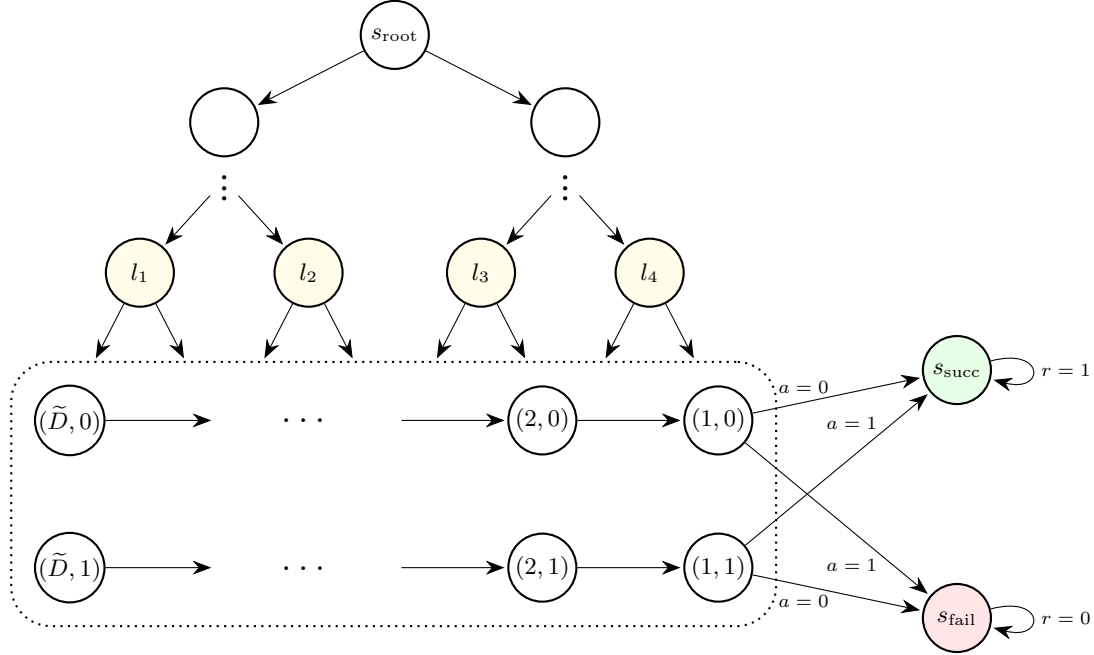


Figure 3. Illustration of hard instances for Theorem 3. The structure consists of a tree structure (top) and a CodeMDP (bottom). The leaf states are labeled l_1, l_2, l_3, l_4 . Each leaf state and action pair has its own probability distribution over the states in the CodeMDP. Once the agent takes an action from the leaf state, it must make $\tilde{D}$ actions without observing which state it has landed in. The agent enters the success state s_{succ} if it landed at state (i, b) and the i -th out of $\tilde{D}$ actions is b , and it receives the reward. The agent enters the fail state s_{fail} if it landed at state (i, b) and the i -th out of $\tilde{D}$ actions is not b , and it cannot receive any reward.

We denote the part that depends on θ by $\tilde{V}^*(\theta)$, defined as

$$\tilde{V}^*(\theta) := \frac{1}{2\tilde{D}} \sum_{i=1}^{\tilde{D}} (\tilde{H} - i) |\theta_i|.$$

Although the optimal value function is no longer a function of $\|\theta\|_1$, we note that this modification does not change the observation type, so the proof of Proposition 2 is still valid, where a slightly modified concentration result for this weighted norm is required. By translating Proposition 2 to the CodeMDP setting, we obtain the following lemma.

Lemma 19. Suppose $\varepsilon \in (0, \frac{1}{2}]$, $\tilde{D}$, and $\tilde{H}$ are given. Assume that $\tilde{D}$ is larger than some absolute constant, $\tilde{H} \geq \frac{H}{2}$, and $\tilde{D} \leq \tilde{H} - 1$. Then, there exist two probability distributions $\mathcal{P}_1$ and $\mathcal{P}_2$ over θ that satisfy the following:

1. For all $n \leq \frac{\tilde{D} \log \tilde{D}}{2\varepsilon^2}$, we have $\text{KL}(\mathcal{P}_1^n \parallel \mathcal{P}_2^n) \leq \frac{1}{8}$, where $\mathcal{P}_1^n, \mathcal{P}_2^n$ denotes the distribution over n samples of $\mathcal{P}_1$ and $\mathcal{P}_2$, respectively.
2. There exists a threshold value α such that $\mathbb{P}_{\theta \sim \mathcal{P}_1}(\tilde{V}^*(\theta) \geq \alpha) \leq \frac{1}{4}$ and $\mathbb{P}_{\theta \sim \mathcal{P}_2}(\tilde{V}^*(\theta) \leq \alpha + \frac{\varepsilon H}{192 \log \tilde{D}}) \leq \frac{1}{4}$.

The proof of Lemma 19 uses the same technical tools as the proof of Proposition 2. The full proof is deferred to Appendix F.1

As we work with parameterized MDPs with randomized parameters, we need a lemma that bounds the KL-divergence of two mixture distributions over the trajectories induced by random MDPs. We define the sample space Ω to be the space of trajectories $\{(s_1^k, a_1^k, \dots, s_H^k, a_H^k, s_{H+1}^k)\}_k$. Assuming a fixed algorithm is given, we denote the probability distribution over Ω induced by an MDP $\mathcal{M}$ over K episodes by $\mathbb{P}_{\mathcal{M}}^K$. We also define $\mathbb{P}_{\mathcal{M}}^\tau$ for a stopping time τ in the same way.

Lemma 20. Let $\mathcal{M}(\theta)$ be an MDP parameterized by $\theta \in \Theta$, where θ only affects the transition distribution of a single state-action pair $(s_0, a_0) \in \mathcal{S} \times \mathcal{A}$. Let $\mathcal{P}_1$ and $\mathcal{P}_2$ be two distributions over Θ with $\mathcal{P}_1 \ll \mathcal{P}_2$. Suppose n and K are given positive integers. Let $\tau \in [K + 1]$ be the index of the episode k at which $N^k(s_0, a_0)$ reaches n , where $\tau = K + 1$ if there is

no such episode. For simplicity, assume that s_0 can be visited at most once per episode for all θ . Assuming the algorithm is fixed, define $\mathbb{P}_1^\tau$ as the mixture distribution of $\mathbb{P}_{\mathcal{M}(\theta)}^\tau$ with $\theta \sim \mathcal{P}_1$, and define $\mathbb{P}_2^\tau$ in the same way with $\theta \sim \mathcal{P}_2$. Then, we have

$$\text{KL}(\mathbb{P}_1^\tau \parallel \mathbb{P}_2^\tau) \leq \text{KL}(\mathcal{P}_1^n \parallel \mathcal{P}_2^n),$$

where $\mathcal{P}_i^n$ is the distribution over n samples of $P_{s_0, a_0}(\theta)$ with $\theta \sim \mathcal{P}_i$ for $i = 1, 2$.

The proof of Lemma 20 is deferred to Appendix F.2.

Now, we prove Theorem 3.

Proof of Theorem 3. Create a combination of a tree-structured MDP and a CodeMDP as described in Appendix D.2. Fix $\varepsilon \in (0, \frac{1}{2}]$, whose value is assigned later. Let $n = \frac{\tilde{D} \log \tilde{D}}{2\varepsilon^2}$.

For the distribution $\mathcal{P}_1$ defined in Lemma 19, we define $\mathcal{P}'_1$ to be its conditional distribution, conditioned on the event $V^*(\theta) < \alpha$, which is well-defined by the same lemma. We set the transition distributions of the leaf-action pairs as $P(l, a) = P(\theta_{l,a})$ with $\theta_{l,a} \stackrel{iid}{\sim} \mathcal{P}'_1$ for all $(l, a) \in \mathcal{L} \times \mathcal{A}$. Let $\mathbb{P}_0$ be the mixture distribution over K trajectories induced by the sampling of an MDP in the described way and then interacting with the given algorithm for K episodes. We denote $\mathbb{E}_0$ as the corresponding expectation.

As we have $\mathbb{E}_0[\sum_{(l,a) \in \mathcal{L} \times \mathcal{A}} N^{K+1}(l, a)] = K$, there exists a leaf-action pair (l_0, a_0) such that $\mathbb{E}_1[N^{K+1}(l_0, a_0)] \leq \frac{K}{L\mathcal{A}}$, where $L := |\mathcal{L}|$. By the Markov inequality, we have $\mathbb{P}_0(N^{K+1}(l_0, a_0) \geq n) \leq \frac{K}{L\mathcal{A}n}$.

Let $\mathbb{P}_1$ and $\mathbb{P}_2$ be mixture distributions that are similar to $\mathbb{P}_0$ except for one difference: the distribution $\mathbb{P}_1$ samples $P(l_0, a_0)$ from $\mathcal{P}_1$, and $\mathbb{P}_2$ samples $P(l_0, a_0)$ from $\mathcal{P}_2$. In addition, we let τ be the index of the episode where $N^\tau(l_0, a_0)$ reaches n for the first time, where $\tau = K + 1$ if there is no such episode. Defining $\mathbb{P}_1^\tau$ and $\mathbb{P}_2^\tau$ as distributions over the first τ episodes under $\mathbb{P}_1$ and $\mathbb{P}_2$ respectively, the KL-divergence between the two distributions is bounded as

$$\begin{aligned} \text{KL}(\mathbb{P}_1^\tau \parallel \mathbb{P}_2^\tau) &\leq \mathbb{E}_{\substack{\theta_{l,a} \sim \mathcal{P}'_1 \\ \forall (l,a) \in \mathcal{L} \times \mathcal{A} \setminus \{(l_0, a_0)\}}} [\text{KL}(\mathbb{P}_1^\tau(\cdot \mid \{\theta_{l,a}\}_{l,a}) \parallel \mathbb{P}_2^\tau(\cdot \mid \{\theta_{l,a}\}_{l,a}))] \\ &\leq \mathbb{E}_{\substack{\theta_{l,a} \sim \mathcal{P}'_1 \\ \forall (l,a) \in \mathcal{L} \times \mathcal{A} \setminus \{(l_0, a_0)\}}} [\text{KL}(\mathcal{P}_1^n \parallel \mathcal{P}_2^n)] \\ &\leq \frac{1}{8}, \end{aligned}$$

where the first inequality is due to the convexity of KL-divergence, the second inequality comes from Lemma 20, and the last inequality applies Lemma 19. Then, by Pinsker's inequality (Lemma 28), we derive that

$$\mathbb{P}_1^\tau(N^\tau(l_0, a_0) \geq n) + \mathbb{P}_2^\tau(N^\tau(l_0, a_0) < n) \geq 1 - \sqrt{\frac{1}{2} \text{KL}(\mathbb{P}_1^\tau \parallel \mathbb{P}_2^\tau)} \geq \frac{3}{4}.$$

We note that $\mathbb{P}_1^\tau(N^\tau(l_0, a_0) \geq n) = \mathbb{P}_1(N^{K+1}(l_0, a_0) \geq n)$ holds by the stopping rule for τ . $\mathbb{P}_2^\tau(N^\tau(l_0, a_0) < n) = \mathbb{P}_2(N^{K+1}(l_0, a_0) < n)$ holds by the same reason. Using that $\mathbb{P}(A \cap B) \geq \mathbb{P}(A) - \mathbb{P}(B^c)$ for any events A and B , we obtain that

$$\begin{aligned} &\mathbb{P}_1 \left(N^{K+1}(l_0, a_0) \geq n, \tilde{V}^*(\theta_{l_0, a_0}) < \alpha \right) \\ &\quad + \mathbb{P}_2 \left(N^{K+1}(l_0, a_0) < n, \tilde{V}^*(\theta_{l_0, a_0}) > \alpha + \frac{\varepsilon H}{192 \log \tilde{D}} \right) \\ &\geq \mathbb{P}_1 \left(N^{K+1}(l_0, a_0) \geq n \right) - \mathbb{P}_1 \left(\tilde{V}^*(\theta_{l_0, a_0}) \geq \alpha \right) \\ &\quad + \mathbb{P}_2 \left(N^{K+1}(l_0, a_0) < n \right) - \mathbb{P}_2 \left(\tilde{V}^*(\theta_{l_0, a_0}) \leq \alpha + \frac{\varepsilon H}{192 \log \tilde{D}} \right) \\ &\geq \mathbb{P}_1 \left(N^{K+1}(l_0, a_0) \geq n \right) + \mathbb{P}_2 \left(N^{K+1}(l_0, a_0) < n \right) - \frac{1}{2} \\ &\geq \frac{1}{4}, \end{aligned}$$

where we apply Lemma 19 for the second inequality. Rearranging the inequality, we have

$$\mathbb{P}_2 \left(N^{K+1}(l_0, a_0) < n, \tilde{V}^*(\boldsymbol{\theta}_{l_0, a_0}) > \alpha + \frac{\varepsilon H}{192 \log \tilde{D}} \right) \geq \frac{1}{4} - \mathbb{P}_1 \left(N^{K+1}(l_0, a_0) \geq n, \tilde{V}^*(\boldsymbol{\theta}_{l_0, a_0}) < \alpha \right). \quad (10)$$

We modify the probability on the right-hand side as follows:

$$\begin{aligned} & \mathbb{P}_1 \left(N^{K+1}(l_0, a_0) \geq n, \tilde{V}^*(\boldsymbol{\theta}_{l_0, a_0}) < \alpha \right) \\ &= \mathbb{P}_1 \left(N^{K+1}(l_0, a_0) \geq n \mid \tilde{V}^*(\boldsymbol{\theta}_{l_0, a_0}) < \alpha \right) \mathbb{P}_1 \left(\tilde{V}^*(\boldsymbol{\theta}_{l_0, a_0}) < \alpha \right) \\ &\leq \frac{3}{4} \mathbb{P}_1 \left(N^{K+1}(l_0, a_0) \geq n \mid \tilde{V}^*(\boldsymbol{\theta}_{l_0, a_0}) < \alpha \right) \\ &= \frac{3}{4} \mathbb{P}_0 \left(N^{K+1}(l_0, a_0) \geq n \right) \\ &\leq \frac{3K}{4LA n}, \end{aligned}$$

where we use Lemma 19 for the first inequality, and the following equality is due to that $\mathbb{P}_0$ is the conditional distribution of $\mathbb{P}_1$ conditioned on the event $\tilde{V}^*(\boldsymbol{\theta}_{l_0, a_0}) < \alpha$. Now, we choose $\varepsilon = \sqrt{\frac{\tilde{D}SA \log \tilde{D}}{96K}}$ so that $n = \frac{\tilde{D} \log \tilde{D}}{2\varepsilon^2} = \frac{48K}{SA} \geq \frac{6K}{LA}$, where we use that $L \geq \frac{S}{8}$ by construction. We note that $\varepsilon < \frac{1}{2}$ is guaranteed by $K \geq \frac{1}{24} \tilde{D}SA \log \tilde{D}$. With this choice of ε , we have $\frac{3K}{4LA n} \leq \frac{1}{8}$. Then, from inequality (10), we obtain that

$$\mathbb{P}_2 \left(N^{K+1}(l_0, a_0) < n, \tilde{V}^*(\boldsymbol{\theta}_{l_0, a_0}) > \alpha + \frac{\varepsilon H}{192 \log \tilde{D}} \right) \geq \frac{1}{8}.$$

It implies that there exists an MDP instance $\mathcal{M}^*$ such that $\tilde{V}^*(\boldsymbol{\theta}_{l_0, a_0}) > \alpha + \frac{\varepsilon H}{192 \log \tilde{D}}$ holds for (l_0, a_0) , and $\tilde{V}^*(\boldsymbol{\theta}_{l, a}) < \alpha$ holds for all $(l, a) \in \mathcal{L} \times \mathcal{A} \setminus \{l_0, a_0\}$, while $\mathbb{P}_{\mathcal{M}^*}(N^{K+1}(l_0, a_0) < n) \geq \frac{1}{8}$ simultaneously holds. For this MDP, taking any leaf-action pair (l, a) other than (l_0, a_0) yields an instantaneous regret of at least $\frac{\varepsilon H}{192 \log \tilde{D}}$. Then, the cumulative regret is bounded below by $\frac{\varepsilon H}{192 \log \tilde{D}} \mathbb{E}_{\mathcal{M}^*}[K - N^{K+1}(l_0, a_0)]$. We lower bound the expected number of sub-optimal leaf-action pair selections as follows:

$$\begin{aligned} \mathbb{E}_{\mathcal{M}^*}[K - N^{K+1}(l_0, a_0)] &\geq \frac{K}{2} \mathbb{P}_{\mathcal{M}^*} \left(K - N^{K+1}(l_0, a_0) \geq \frac{K}{2} \right) \\ &= \frac{K}{2} \mathbb{P}_{\mathcal{M}^*} \left(\frac{K}{2} \geq N^{K+1}(l_0, a_0) \right) \\ &\geq \frac{K}{2} \mathbb{P}_{\mathcal{M}^*} (n \geq N^{K+1}(l_0, a_0)) \\ &\geq \frac{K}{16}, \end{aligned}$$

where the first inequality uses Markov's inequality, and the third line uses that $n = \frac{K}{4SA} < \frac{K}{2}$. Therefore, the cumulative regret for $\mathcal{M}^*$ is lower bounded by

$$\begin{aligned} \frac{\varepsilon H}{192 \log \tilde{D}} \mathbb{E}_{\mathcal{M}^*}[K - N^{K+1}(l_0, a_0)] &\geq \frac{\varepsilon H K}{3072 \log \tilde{D}} \\ &= \frac{H}{3072} \sqrt{\frac{\tilde{D}SAK}{96 \log \tilde{D}}}. \end{aligned}$$

The proof is complete. $\square$

E. Proof of Proposition 1

In this section, we focus on the ℓ_1 -estimation problem introduced in Appendix D.1. Proposition 2, which is the formal version of Proposition 1, states that the lower bound for the sample complexity is $\Omega(\frac{d}{\varepsilon^2 \log d})$. The main ideas presented in this section to prove Proposition 2 also apply to the proof of Lemma 19, which is a core lemma in proving Theorem 3.

The proof of Proposition 2 is based on the method of two fuzzy hypotheses (Tsybakov, 2008). The two distributions over θ are based on the distributions introduced in Cai & Low (2011).

Lemma 21 (Lemma 1 in Cai & Low (2011)). *For given even integer $k > 0$, there exist two probability measures ν_1 and ν_2 on $[-1, 1]$ that satisfy the following conditions:*

- ν_1 and ν_2 are symmetric around 0.
- $\int t^l d\nu_1 = \int t^l d\nu_2$ for $l = 0, \dots, k$.
- $\int |t| d\nu_2 - \int |t| d\nu_1 = 2\delta_k$, where $\delta_k := \inf_{p \in \mathcal{P}_k} \sup_{x \in [-1, 1]} ||x| - p(x)| \geq \frac{1}{4k}$.

We denote $\mu_1 := \int |t| d\nu_1$ and $\mu_2 := \int |t| d\nu_2$. By Lemma 21, we have $\mu_2 - \mu_1 \geq \frac{1}{2k}$. Additionally, for a real number $\varepsilon > 0$, we denote the distribution of εZ_i with $Z_i \sim \rho$ by $\varepsilon \rho$ for simplicity.

With this distribution, we can prove the following KL-divergence bound on the two fuzzy hypotheses.

Proposition 3. *For a given even integer $k > 0$, let ν_1 and ν_2 be defined as in Lemma 21. For a given $\varepsilon \in (0, \frac{1}{2}]$, suppose a prior distribution $\mathcal{P}_1$ on $\{\theta_i\}_{i=1}^d$ is given as $\theta_i \stackrel{i.i.d.}{\sim} \varepsilon \nu_1$, and $\mathcal{P}_2$ in the same way with ν_2 . Let $\mathcal{P}_1^n$ and $\mathcal{P}_2^n$ be the probability measures on n observed samples of $\{(I_t, X_t)\}_{t=1}^n$ with given priors. Then,*

$$\text{KL}(\mathcal{P}_1^n \parallel \mathcal{P}_2^n) \leq d \log \left(1 + \left(\frac{4n\varepsilon^2}{d(k+1)} \right)^{k+1} \left(1 + \frac{2\varepsilon^2}{d} \right)^n \right).$$

In particular, when $n \leq \frac{d \log d}{2\varepsilon^2}$ and $k \geq 4 \log d + 2$, one has $\text{KL}(\mathcal{P}_1^n \parallel \mathcal{P}_2^n) \leq \frac{1}{8}$.

Proof of Proposition 3 is deferred to Appendix E.1.

Proof of Proposition 2. We assume that d is larger than some absolute constant which we specify later, and $\varepsilon \leq \frac{1}{48 \log d}$. Let $n = \frac{d}{1152\varepsilon^2 \log d}$. We show that for any estimator $\hat{L}^n$, there exists an instance θ that satisfies $\mathbb{P}_\theta(|\hat{L}^n - \frac{1}{d} \|\theta\|_1| \geq \varepsilon) \geq \frac{1}{8}$. Let $\mathbb{P}_1 := \mathcal{P}_1^n$ and $\mathbb{P}_2 = \mathcal{P}_2^n$ be probability distributions defined in Proposition 3 with k, ε' , and n . We plug in $k = 6 \log d \geq 4 \log d + 2$, $\varepsilon' = 4k\varepsilon$, where we assume that $d \geq 3$. Note that $n = \frac{d}{1152\varepsilon^2 \log d} \leq \frac{d \log d}{2(\varepsilon')^2}$, so by Proposition 3 we have we have $\text{KL}(\mathbb{P}_1 \parallel \mathbb{P}_2) \leq \frac{1}{8}$.

Let $\psi = \mathbb{1}\{\hat{L}^n \leq \frac{\varepsilon'(\mu_1 + \mu_2)}{2}\}$, where μ_1 and μ_2 are defined after Lemma 21. By Pinsker's inequality (Lemma 28), we have

$$\mathbb{P}_1(\psi = 0) + \mathbb{P}_2(\psi = 1) \geq 1 - \sqrt{\frac{1}{2} \text{KL}(\mathbb{P}_1 \parallel \mathbb{P}_2)} \geq \frac{3}{4}. \quad (11)$$

Using Hoeffding's inequality (Lemma 27), we have

$$\begin{aligned} \mathbb{P}_1 \left(\frac{1}{d} \|\theta\|_1 > \varepsilon' \left(\mu_1 + \frac{1}{48 \log d} \right) \right) &= \mathbb{P}_1 \left(\frac{1}{d} \sum_{i=1}^d (|\theta_i| - \varepsilon' \mu_1) > \frac{\varepsilon'}{48 \log d} \right) \\ &\leq \exp \left(-\frac{d}{1152(\log d)^2} \right). \end{aligned}$$

We assume that the value of d is larger than some absolute constant, so that $\mathbb{P}_1 \left(\frac{1}{d} \|\theta\|_1 > \varepsilon' \left(\mu_1 + \frac{1}{48 \log d} \right) \right) \leq \frac{1}{4}$. We note that Lemma 21 implies that $\mu_2 - \mu_1 \geq \frac{1}{2k} \geq \frac{1}{12 \log d}$, so

$$\begin{aligned} \varepsilon' \left(\mu_1 + \frac{1}{48 \log d} \right) &\leq \varepsilon' \left(\mu_1 + \frac{\mu_2 - \mu_1}{2} - \frac{1}{48 \log d} \right) \\ &= \frac{\varepsilon'(\mu_1 + \mu_2)}{2} - \varepsilon, \end{aligned}$$

which implies that $\mathbb{P}_1(\frac{1}{d}\|\boldsymbol{\theta}\|_1 > \frac{\varepsilon'(\mu_1+\mu_2)}{2} - \varepsilon) \leq \frac{1}{4}$. In the same way, we obtain that $\mathbb{P}_2(\frac{1}{d}\|\boldsymbol{\theta}\|_1 < \frac{\varepsilon'(\mu_1+\mu_2)}{2} + \varepsilon) \leq \frac{1}{4}$. Using these bounds and inequality (11), we have

$$\begin{aligned} & \mathbb{P}_1\left(\psi = 0, \frac{1}{d}\|\boldsymbol{\theta}\|_1 \leq \frac{\varepsilon'(\mu_1+\mu_2)}{2} - \varepsilon\right) + \mathbb{P}_2\left(\psi = 1, \frac{1}{d}\|\boldsymbol{\theta}\|_1 \geq \frac{\varepsilon'(\mu_1+\mu_2)}{2} + \varepsilon\right) \\ & \geq \mathbb{P}_1(\psi = 0) - \mathbb{P}_1\left(\frac{1}{d}\|\boldsymbol{\theta}\|_1 > \frac{\varepsilon'(\mu_1+\mu_2)}{2} - \varepsilon\right) \\ & \quad + \mathbb{P}_2(\psi = 1) - \mathbb{P}_2\left(\frac{1}{d}\|\boldsymbol{\theta}\|_1 < \frac{\varepsilon'(\mu_1+\mu_2)}{2} + \varepsilon\right) \\ & \geq \frac{1}{4}. \end{aligned} \tag{12}$$

Note that the events in the first line are subsets of the event that the estimator $\hat{L}^n$ is wrong by a difference of at least ε . Therefore, inequality (12) shows that there must exists an instance $\boldsymbol{\theta}$ where $\mathbb{P}_{\boldsymbol{\theta}}(|\hat{L}^n - \frac{1}{d}\|\boldsymbol{\theta}\|_1| \geq \varepsilon) \geq \frac{1}{8}$. $\square$

E.1. Proof of Proposition 3

Proof of Proposition 3. Let Ω be the sample space of all possible sequence of $\{(I_t, X_t)\}_{t=1}^n$. Let $\omega = \{I_t, X_t\}_{t=1}^n$. Then, we have that

$$\begin{aligned} \mathcal{P}_1^n(\omega) &= \mathbb{E}_{\theta_i \stackrel{i.i.d.}{\sim} \varepsilon\nu_1} \left[\prod_{t=1}^n \left(\frac{1 + (-1)^{X_t+1}\theta_{I_t}}{2d} \right) \right] \\ &= \left(\frac{1}{2d} \right)^n \mathbb{E}_{\theta_i \stackrel{i.i.d.}{\sim} \varepsilon\nu_1} \left[\prod_{i=1}^d (1 - \theta_i)^{N(i,0)} (1 + \theta_i)^{N(i,1)} \right], \end{aligned}$$

where we define $N(i, 0) := N(i, 0, \omega) = \sum_{t=1}^n \mathbb{1}\{I_t = i, X_t = 0\}$ and $N(i, 1)$ in the same way for $X_t = 1$. For $x \in \mathbb{R}$ and $a, b \in \mathbb{N} \cup \{0\}$, define $f(x, a, b) := (1-x)^a(1+x)^b$. Then, the probability law is expressed as

$$\mathcal{P}_1^n(\omega) = \left(\frac{1}{2d} \right)^n \prod_{i=1}^d \mathbb{E}_{\theta_i \stackrel{i.i.d.}{\sim} \varepsilon\nu_1} [f(\theta_i, N(i, 0), N(i, 1))],$$

where we use the fact that θ_i are mutually independent of one another. The expression holds for $\mathcal{P}_2^n$ with ν_2 instead of ν_1 . The KL divergence between $\mathcal{P}_1^n$ and $\mathcal{P}_2^n$ becomes

$$\begin{aligned} \text{KL}(\mathcal{P}_1^n \parallel \mathcal{P}_2^n) &= \mathbb{E}_{\mathcal{P}_1^n} \left[\log \frac{d\mathcal{P}_1^n}{d\mathcal{P}_2^n} \right] \\ &= \mathbb{E}_{\mathcal{P}_1^n} \left[\log \prod_{i=1}^d \frac{\mathbb{E}_{\theta_i' \stackrel{i.i.d.}{\sim} \varepsilon\nu_1} [f(\theta_i', N(i, 0), N(i, 1))]}{\mathbb{E}_{\theta_i' \stackrel{i.i.d.}{\sim} \varepsilon\nu_2} [f(\theta_i', N(i, 0), N(i, 1))]} \right] \\ &= \mathbb{E}_{\mathcal{P}_1^n} \left[\sum_{i=1}^d \log \frac{\mathbb{E}_{\theta_i' \stackrel{i.i.d.}{\sim} \varepsilon\nu_1} [f(\theta_i', N(i, 0), N(i, 1))]}{\mathbb{E}_{\theta_i' \stackrel{i.i.d.}{\sim} \varepsilon\nu_2} [f(\theta_i', N(i, 0), N(i, 1))]} \right] \\ &= \mathbb{E}_{\mathcal{P}_1^n} \left[\sum_{i=1}^d \log \frac{\mathbb{E}_{\theta' \sim \varepsilon\nu_1} [f(\theta', N(i, 0), N(i, 1))]}{\mathbb{E}_{\theta' \sim \varepsilon\nu_2} [f(\theta', N(i, 0), N(i, 1))]} \right]. \end{aligned}$$

Note that $(N(i, 0), N(i, 1))$ have the same distribution across indices $i = 1, \dots, d$. We define random variables N_0 and N_1 that have the same distribution as $N(i, 0)$ and $N(i, 1)$, respectively. Under $\mathcal{P}_1^n$, the distributions of N_0 and N_1 follow $B(n, \frac{1-\theta}{2d})$ and $B(n, \frac{1+\theta}{2d})$, respectively, where $\theta \sim \varepsilon\nu_1$. Then, we can express the likelihood ratio for each index in the same form and obtain

$$\text{KL}(\mathcal{P}_1^n \parallel \mathcal{P}_2^n) = d \mathbb{E}_{\mathcal{P}_1^n} \left[\log \frac{\mathbb{E}_{\theta' \sim \varepsilon\nu_1} [f(\theta', N_0, N_1) \mid N_0, N_1]}{\mathbb{E}_{\theta' \sim \varepsilon\nu_2} [f(\theta', N_0, N_1) \mid N_0, N_1]} \right].$$

This can be understood as that the KL divergence between $\mathcal{P}_1^n$ and $\mathcal{P}_2^n$ is d times the KL divergence between the probability measures restricted on one index. Using that $\text{KL}(P||Q) \leq \log(1 + \chi^2(P; Q))$, where $\chi^2(P; Q)$ is the chi-squared divergence, we have

$$\begin{aligned} & \mathbb{E}_{(N_0, N_1) \sim \mathcal{P}_1^n} \left[\log \frac{\mathbb{E}_{\theta' \sim \varepsilon \nu_1} [f(\theta', N_0, N_1)]}{\mathbb{E}_{\theta' \sim \varepsilon \nu_2} [f(\theta', N_0, N_1)]} \right] \\ & \leq \log \left(1 + \mathbb{E}_{(N_0, N_1) \sim \mathcal{P}_2^n} \left[\left(\frac{\mathbb{E}_{\theta' \sim \varepsilon \nu_1} [f(\theta', N_0, N_1)] - \mathbb{E}_{\theta' \sim \varepsilon \nu_2} [f(\theta', N_0, N_1)]}{\mathbb{E}_{\theta' \sim \varepsilon \nu_2} [f(\theta', N_0, N_1)]} \right)^2 \right] \right), \end{aligned}$$

where we omit conditioning on N_0, N_1 for the inner expectations for simplicity. For the ease of presentation, let $\mathcal{P}_0^n$ be the probability distribution over N_0 and N_1 when $\theta = 0$ deterministically. Then, we have $\frac{d\mathcal{P}_2^n}{d\mathcal{P}_0^n} = \frac{\mathbb{E}_{\theta' \sim \varepsilon \nu_2} [f(\theta', N_0, N_1)]}{f(0, N_0, N_1)} = \mathbb{E}_{\theta' \sim \varepsilon \nu_2} [f(\theta', N_0, N_1)]$. Then, the chi-squared divergence can be expressed as

$$\begin{aligned} \chi^2 &:= \mathbb{E}_{(N_0, N_1) \sim \mathcal{P}_2^n} \left[\left(\frac{\mathbb{E}_{\theta' \sim \varepsilon \nu_1} [f(\theta', N_0, N_1)] - \mathbb{E}_{\theta' \sim \varepsilon \nu_2} [f(\theta', N_0, N_1)]}{\mathbb{E}_{\theta' \sim \varepsilon \nu_2} [f(\theta', N_0, N_1)]} \right)^2 \right] \\ &= \mathbb{E}_{(N_0, N_1) \sim \mathcal{P}_0^n} \left[\frac{d\mathcal{P}_2^n}{d\mathcal{P}_0^n} \left(\frac{\mathbb{E}_{\theta' \sim \varepsilon \nu_1} [f(\theta', N_0, N_1)] - \mathbb{E}_{\theta' \sim \varepsilon \nu_2} [f(\theta', N_0, N_1)]}{\mathbb{E}_{\theta' \sim \varepsilon \nu_2} [f(\theta', N_0, N_1)]} \right)^2 \right] \\ &= \mathbb{E}_{(N_0, N_1) \sim \mathcal{P}_0^n} \left[\frac{(\mathbb{E}_{\theta' \sim \varepsilon \nu_1} [f(\theta', N_0, N_1)] - \mathbb{E}_{\theta' \sim \varepsilon \nu_2} [f(\theta', N_0, N_1)])^2}{\mathbb{E}_{\theta' \sim \varepsilon \nu_2} [f(\theta', N_0, N_1)]} \right] \end{aligned}$$

We first lower bound the denominator. Since ν_2 is symmetric, we have

$$\mathbb{E}_{\theta' \sim \varepsilon \nu_2} [f(\theta', N_0, N_1)] = \mathbb{E}_{\theta' \sim \varepsilon \nu_2} [f(-\theta', N_0, N_1)] = \frac{1}{2} \mathbb{E}_{\theta' \sim \varepsilon \nu_2} [f(\theta', N_0, N_1) + f(-\theta', N_0, N_1)].$$

Using the AM-GM inequality, we obtain that

$$\begin{aligned} \frac{1}{2} (f(\theta', N_0, N_1) + f(-\theta', N_0, N_1)) &= \frac{1}{2} ((1 - \theta')^{N_0} (1 + \theta')^{N_1} + (1 + \theta')^{N_0} (1 - \theta')^{N_1}) \\ &\geq ((1 + \theta')^{N_0 + N_1} (1 - \theta')^{N_0 + N_1})^{\frac{1}{2}} \\ &= (1 - \theta'^2)^{\frac{N_0 + N_1}{2}} \\ &\geq (1 - \varepsilon^2)^{\frac{N_0 + N_1}{2}}, \end{aligned}$$

where the last inequality uses that $\theta' \in [-\varepsilon, \varepsilon]$. Therefore, denoting $N := N_0 + N_1$, the chi-squared divergence is upper bounded by

$$\chi^2 \leq \mathbb{E}_{(N_0, N_1) \sim \mathcal{P}_0^n} \left[(1 - \varepsilon^2)^{-\frac{N}{2}} \left(\mathbb{E}_{\theta' \sim \varepsilon \nu_1} [f(\theta', N_0, N_1)] - \mathbb{E}_{\theta' \sim \varepsilon \nu_2} [f(\theta', N_0, N_1)] \right)^2 \right].$$

Conditioned on N , the conditional distribution of N_0 under $\mathcal{P}_0^n$ follows $B(N, \frac{1}{2})$. Hence, using the law of total expectation, we can rewrite the bound as

$$\chi^2 \leq \mathbb{E}_{N \sim \mathcal{P}_0^n} \left[(1 - \varepsilon^2)^{-\frac{N}{2}} \mathbb{E}_{\substack{N_0 \sim B(N, \frac{1}{2}) \\ N_1 = N - N_0}} \left[\left(\mathbb{E}_{\theta' \sim \varepsilon \nu_1} [f(\theta', N_0, N_1)] - \mathbb{E}_{\theta' \sim \varepsilon \nu_2} [f(\theta', N_0, N_1)] \right)^2 \middle| N \right] \right]. \quad (13)$$

We transform the inner expectation using the following lemma:

Lemma 22. For $N \in \mathbb{N}$, let $N_0 \sim B(N, \frac{1}{2})$ and $N_1 = N - N_0$. Recall that $f(x; a, b) := (1 - x)^a (1 + x)^b$. For any two distributions ρ_1 and ρ_2 independent of N_0 , we have

$$\mathbb{E}_{N_0, N_1} \left[\left(\mathbb{E}_{Y \sim \rho_1} [f(Y, N_0, N_1)] - \mathbb{E}_{Y \sim \rho_2} [f(Y, N_0, N_1)] \right)^2 \right] = \sum_{l=0}^N \binom{N}{l} \left(\mathbb{E}_{Y \sim \rho_1} [Y^l] - \mathbb{E}_{Y \sim \rho_2} [Y^l] \right)^2.$$

The proof of Lemma 22 is provided in Appendix F.3 By applying Lemma 22, we have

$$\begin{aligned}
 & \mathbb{E}_{\substack{N_0 \sim B(N, \frac{1}{2}) \\ N_1 = N - N_0}} \left[\left(\mathbb{E}_{\theta' \sim \varepsilon \nu_1} [f(\theta', N_0, N_1)] - \mathbb{E}_{\theta' \sim \varepsilon \nu_2} [f(\theta', N_0, N_1)] \right)^2 \middle| N \right] \\
 &= \sum_{l=0}^N \binom{N}{l} \left(\mathbb{E}_{\theta' \sim \varepsilon \nu_1} [(\theta')^l] - \mathbb{E}_{\theta' \sim \varepsilon \nu_2} [(\theta')^l] \right)^2 \\
 &= \sum_{l=k+1}^N \binom{N}{l} \left(\mathbb{E}_{\theta' \sim \varepsilon \nu_1} [(\theta')^l] - \mathbb{E}_{\theta' \sim \varepsilon \nu_2} [(\theta')^l] \right)^2 \\
 &\leq \sum_{l=k+1}^N \binom{N}{l} \varepsilon^{2l},
 \end{aligned}$$

where we use the property $\mathbb{E}_{\theta' \sim \varepsilon \nu_1} [(\theta')^l] = \mathbb{E}_{\theta' \sim \varepsilon \nu_2} [(\theta')^l]$ for $l = 0, \dots, k$ by Lemma 21 for the second equality, and the last inequality uses that $|\theta'| \leq \varepsilon$. Plugging this result into Eq. (13), we obtain that

$$\chi^2 \leq \mathbb{E}_{N \sim \mathcal{P}_0^n} \left[(1 - \varepsilon^2)^{-\frac{N}{2}} \sum_{l=k+1}^N \binom{N}{l} \varepsilon^{2l} \right].$$

Then, we apply Taylor's theorem, whose result is encapsulated by the following lemma:

Lemma 23. *For $n \in \mathbb{N}$, $p \in [0, 1]$, and $a, x \in \mathbb{R}$, there exists $\tau \in [0, 1]$ such that*

$$\mathbb{E}_{N \sim B(n, p)} \left[a^N \sum_{l=k+1}^N \binom{N}{l} x^l \right] = \binom{n}{k+1} (apx)^{k+1} (ap(1 + \tau x) + 1 - p)^{n-k-1}.$$

The proof of Lemma 23 is provided in Appendix F.4. Recall that $N \sim B(n, \frac{1}{d})$ under $\mathcal{P}_0^n$. Plugging in $x = \varepsilon^2$, $a = \frac{1}{\sqrt{1-\varepsilon^2}}$, and $p = \frac{1}{d}$, we obtain that

$$\begin{aligned}
 \chi^2 &\leq \binom{n}{k+1} \left(\frac{\varepsilon^2}{d\sqrt{1-\varepsilon^2}} \right)^{k+1} \left(\frac{1}{d\sqrt{1-\varepsilon^2}} (1 + \tau\varepsilon^2) + 1 - \frac{1}{d} \right)^{n-k-1} \\
 &\leq \left(\frac{en\varepsilon^2}{d(k+1)\sqrt{1-\varepsilon^2}} \right)^{k+1} \left(\frac{1}{d\sqrt{1-\varepsilon^2}} (1 + \tau\varepsilon^2) + 1 - \frac{1}{d} \right)^{n-k-1} \\
 &\leq \left(\frac{en\varepsilon^2}{d(k+1)\sqrt{1-\varepsilon^2}} \right)^{k+1} \left(1 + \frac{1}{d} \left(\frac{1+\varepsilon^2}{\sqrt{1-\varepsilon^2}} - 1 \right) \right)^{n-k-1},
 \end{aligned}$$

where we use $\binom{n}{k+1} \leq \left(\frac{en}{k+1} \right)^{k+1}$ for the second inequality. From $0 \leq \varepsilon \leq \frac{1}{2}$, we bound $\frac{e}{\sqrt{1-\varepsilon^2}}$ in the first term by 4, and $\frac{1+\varepsilon^2}{\sqrt{1-\varepsilon^2}} = \sqrt{\frac{1+2\varepsilon^2+\varepsilon^4}{1-\varepsilon^2}} = \sqrt{1 + \frac{3\varepsilon^2+\varepsilon^4}{1-\varepsilon^2}} \leq \sqrt{1 + 4\varepsilon^2 + \frac{4}{3}\varepsilon^4} \leq 1 + 2\varepsilon^2$. Then, we have

$$\chi^2 \leq \left(\frac{4n\varepsilon^2}{d(k+1)} \right)^{k+1} \left(1 + \frac{2\varepsilon^2}{d} \right)^n,$$

and the KL divergence is bounded by

$$\begin{aligned}
 \text{KL}(\mathcal{P}_1^n \parallel \mathcal{P}_2^n) &\leq d \log(1 + \chi^2) \\
 &\leq d \log \left(1 + \left(\frac{4n\varepsilon^2}{d(k+1)} \right)^{k+1} \left(1 + \frac{2\varepsilon^2}{d} \right)^n \right).
 \end{aligned}$$

The first part of the lemma is proved.

Suppose $n \leq \frac{d \log d}{2\varepsilon^2}$. Then, we have

$$\begin{aligned}\chi^2 &\leq \left(\frac{2 \log d}{k+1}\right)^{k+1} \left(1 + \frac{2\varepsilon^2}{d}\right)^{\frac{d \log d}{2\varepsilon^2}} \\ &\leq \left(\frac{2 \log d}{k+1}\right)^{k+1} e^{\log d} \\ &= \left(\frac{2 \log d}{k+1}\right)^{k+1} d.\end{aligned}$$

By setting $k \geq 4 \log d + 2$, we have

$$\left(\frac{2 \log d}{k+1}\right)^{k+1} \leq \left(\frac{1}{2}\right)^{4 \log d + 3} \leq \frac{1}{8} e^{(\log \frac{1}{2}) \cdot (4 \log d)} \leq \frac{1}{8} e^{-2 \log d} \leq \frac{1}{8d^2}.$$

Therefore, we have $\chi^2 \leq \frac{1}{8d}$, and $\text{KL}(\mathcal{P}_1^n \parallel \mathcal{P}_2^n) \leq \log(1 + \chi^2)^d \leq \log(1 + \frac{1}{8d})^d \leq \log e^{1/8} = \frac{1}{8}$. $\square$

F. Proofs of Technical Lemmas in Appendices D and E

F.1. Proof of Lemma 19

Proof of Lemma 19. We set $\mathcal{P}_1$ and $\mathcal{P}_2$ to be distributions defined in Proposition 3 with $d = \tilde{D}$ and $k = 6 \log \tilde{D}$. Then, Proposition 3 immediately yields that $\text{KL}(\mathcal{P}_1^n \parallel \mathcal{P}_2^n) \leq \frac{1}{8}$ for $n \leq \frac{\tilde{D} \log \tilde{D}}{2\varepsilon^2}$.

Now, we prove the second part of the lemma. We define μ_1 and μ_2 as defined after Lemma 21. Specifically, they satisfy $\varepsilon \mu_1 = \mathbb{E}_{\theta \sim \mathcal{P}_1}[|\theta_i|]$ and $\varepsilon \mu_2 = \mathbb{E}_{\theta \sim \mathcal{P}_2}[|\theta_i|]$ for all $i \in [\tilde{D}]$. In addition, Lemma 21 states that $\mu_2 - \mu_1 \geq \frac{1}{2k} = \frac{1}{12 \log \tilde{D}}$. We set $\alpha := \varepsilon(\mu_1 + \frac{1}{48 \log \tilde{D}})^{\frac{2\tilde{H}-1-\tilde{D}}{4}}$. Applying Hoeffding's inequality (Lemma 27), we have

$$\begin{aligned}\mathbb{P}_{\theta \sim \mathcal{P}_1}(\tilde{V}^*(\theta) \geq \alpha) &= \mathbb{P}_{\theta \sim \mathcal{P}_1}\left(\frac{1}{2\tilde{D}} \sum_{i=1}^{\tilde{D}} (\tilde{H} - i)|\theta_i| \geq \varepsilon \left(\mu_1 + \frac{1}{48 \log \tilde{D}}\right) \frac{2\tilde{H} - 1 - \tilde{D}}{4}\right) \\ &= \mathbb{P}_{\theta \sim \mathcal{P}_1}\left(\frac{1}{2\tilde{D}} \sum_{i=1}^{\tilde{D}} (\tilde{H} - i)(|\theta_i| - \varepsilon \mu_1) \geq \frac{\varepsilon}{48 \log \tilde{D}} \cdot \frac{2\tilde{H} - 1 - \tilde{D}}{4}\right) \\ &\leq \exp\left(-\left(\frac{\varepsilon}{48 \log \tilde{D}} \cdot \frac{2\tilde{H} - 1 - \tilde{D}}{4}\right)^2 \cdot \left(\sum_{i=1}^{\tilde{D}} \left(\frac{\varepsilon(\tilde{H} - i)}{2\tilde{D}}\right)^2\right)^{-1}\right) \\ &\leq \exp\left(-\left(\frac{1}{48 \log \tilde{D}} \cdot \frac{\tilde{H}}{4}\right)^2 \cdot \left(\frac{\tilde{H}^2}{4\tilde{D}}\right)^{-1}\right) \\ &= \exp\left(-\frac{\tilde{D}}{(96 \log \tilde{D})^2}\right),\end{aligned}$$

where we use that $\tilde{H} \geq \tilde{D} + 1$ in the fifth line. We note that for large enough $\tilde{D}$, we have $\exp(-\tilde{D}/(96 \log \tilde{D})^2) \leq \frac{1}{4}$. Then we obtain that $\mathbb{P}_{\theta \sim \mathcal{P}_1}(\tilde{V}^*(\theta) \geq \alpha) \leq \frac{1}{4}$.

It remains to show that $\mathbb{P}_{\theta \sim \mathcal{P}_2}(\tilde{V}^*(\theta) \leq \alpha + \frac{\varepsilon \tilde{H}}{128 \log \tilde{D}}) \leq \frac{1}{4}$. We define $\alpha_2 := \varepsilon(\mu_2 - \frac{1}{48 \log \tilde{D}})^{\frac{2\tilde{H}-1-\tilde{D}}{4}}$, and in the exact

same way, we can prove $\mathbb{P}_{\theta \sim \mathcal{P}_2}(\tilde{V}^*(\theta) \leq \alpha_2) \leq \frac{1}{4}$. We have

$$\begin{aligned} \alpha_2 - \alpha &= \varepsilon \left(\mu_2 - \mu_1 - \frac{1}{24 \log \tilde{D}} \right) \frac{2\tilde{H} - 1 - \tilde{D}}{4} \\ &\geq \frac{\varepsilon}{24 \log \tilde{D}} \cdot \frac{2\tilde{H} - 1 - \tilde{D}}{4} \\ &\geq \frac{\varepsilon}{24 \log \tilde{D}} \cdot \frac{\tilde{H}}{4} \\ &\geq \frac{\varepsilon H}{192 \log \tilde{D}}, \end{aligned}$$

where we use $\mu_2 - \mu_1 \geq \frac{1}{12 \log \tilde{D}}$ for the first inequality, the $\tilde{D} + 1 \leq \tilde{H}$ for the second inequality, and $\tilde{H} \geq \frac{H}{2}$ for the last inequality. Therefore, we deduce that $\mathbb{P}_{\theta \sim \mathcal{P}_2}(\tilde{V}^*(\theta) \leq \alpha + \frac{\varepsilon H}{192 \log \tilde{D}}) \leq \mathbb{P}_{\theta \sim \mathcal{P}_2}(\tilde{V}^*(\theta) \leq \alpha_2) \leq \frac{1}{4}$. $\square$

F.2. Proof of Lemma 20

Proof of Lemma 20. We denote $\mathbb{P}_1 := \mathbb{P}_1^\tau$ and $\mathbb{P}_2 := \mathbb{P}_2^\tau$ for simplicity.

For $\omega = \{(s_1^k, a_1^k, \dots, s_{H+1}^k)\}_{k=1}^\tau \in \Omega$, we have

$$\mathbb{P}_i(\omega) = \mathbb{E}_{\substack{\mathcal{M}(\theta) \\ \theta \sim \mathcal{P}_i}} \left[\prod_{k=1}^\tau \prod_{h=1}^H P_{s_h^k, a_h^k}(s_{h+1}^k) \right]$$

for $i = 1$ and $i = 2$. Denote the state sampled from the j -th selection of (s_0, a_0) in the trajectory ω by $s^j(\omega)$. Then, we have that

$$\begin{aligned} \frac{\mathbb{P}_1(\omega)}{\mathbb{P}_2(\omega)} &= \frac{\mathbb{E}_{\mathcal{P}_1} \left[\prod_{\substack{(k,h) \in [\tau] \times [H] \\ (s_h^k, a_h^k) = (s_0, a_0)}} P_{s_0, a_0}(s_{h+1}^k) \right]}{\mathbb{E}_{\mathcal{P}_2} \left[\prod_{\substack{(k,h) \in [\tau] \times [H] \\ (s_h^k, a_h^k) = (s_0, a_0)}} P_{s_0, a_0}(s_{h+1}^k) \right]} \\ &= \frac{\mathbb{E}_{\mathcal{P}_1} \left[\prod_{j=1}^{N^{\tau+1}(s_0, a_0)} P_{s_0, a_0}(s^j(\omega)) \right]}{\mathbb{E}_{\mathcal{P}_2} \left[\prod_{j=1}^{N^{\tau+1}(s_0, a_0)} P_{s_0, a_0}(s^j(\omega)) \right]} \\ &= \frac{\mathcal{P}_1^{N^{\tau+1}}(\{s^j(\omega)\}_{j=1}^{N^{\tau+1}(s_0, a_0)})}{\mathcal{P}_2^{N^{\tau+1}}(\{s^j(\omega)\}_{j=1}^{N^{\tau+1}(s_0, a_0)})}. \end{aligned}$$

The likelihood ratio only depends on the sequence of states that the state-action pair (s_0, a_0) transitioned into, and it is the ratio between $\mathcal{P}_1^{N^{\tau+1}(s_0, a_0)}$ and $\mathcal{P}_2^{N^{\tau+1}(s_0, a_0)}$. Then, the KL-divergence is expressed as the following:

$$\begin{aligned} \text{KL}(\mathbb{P}_1 \parallel \mathbb{P}_2) &= \mathbb{E}_{\omega \sim \mathbb{P}_1} \left[\log \frac{\mathbb{P}_1(\omega)}{\mathbb{P}_2(\omega)} \right] \\ &= \mathbb{E}_{\omega \sim \mathbb{P}_1} \left[\log \frac{\mathcal{P}_1^{N^{\tau+1}}(\{s^j(\omega)\}_{j=1}^{N^{\tau+1}(s_0, a_0)})}{\mathcal{P}_2^{N^{\tau+1}}(\{s^j(\omega)\}_{j=1}^{N^{\tau+1}(s_0, a_0)})} \right] \\ &= \mathbb{E}_{N^{\tau+1}(s_0, a_0) \sim \mathbb{P}_1} \left[\mathbb{E}_{\{s^j(\omega)\}_{j=1}^{N^{\tau+1}(s_0, a_0)}} \left[\log \frac{\mathcal{P}_1^{N^{\tau+1}}(\{s^j(\omega)\}_{j=1}^{N^{\tau+1}(s_0, a_0)})}{\mathcal{P}_2^{N^{\tau+1}}(\{s^j(\omega)\}_{j=1}^{N^{\tau+1}(s_0, a_0)})} \middle| N^{\tau+1}(s_0, a_0) \right] \right] \\ &= \mathbb{E}_{N^{\tau+1}(s_0, a_0) \sim \mathbb{P}_1} \left[\text{KL}(\mathcal{P}_1^{N^{\tau+1}(s_0, a_0)}(\cdot \mid N^{\tau+1}(s_0, a_0)) \parallel \mathcal{P}_2^{N^{\tau+1}(s_0, a_0)}(\cdot \mid N^{\tau+1}(s_0, a_0))) \right]. \end{aligned}$$

By the data processing inequality, we have

$$\text{KL}(\mathcal{P}_1^{N^{\tau+1}(s_0, a_0)}(\cdot \mid N^{\tau+1}(s_0, a_0)) \parallel \mathcal{P}_2^{N^{\tau+1}(s_0, a_0)}(\cdot \mid N^{\tau+1}(s_0, a_0))) \leq \text{KL}(\mathcal{P}_1^n \parallel \mathcal{P}_2^n),$$

where we use that $N^{\tau+1}(s_0, a_0) \leq n$ by the stopping rule of τ . The proof is complete. $\square$

F.3. Proof of Lemma 22

Proof of Lemma 22. Let $c(a, b, l)$ be the coefficients of $f(x, a, b)$ so that $f(x, a, b) = (1-x)^a(1+x)^b = \sum_{l=0}^{a+b} c(a, b, l)x^l$. Then, for given N_0, N_1 , the squared term becomes

$$\begin{aligned} & \left(\mathbb{E}_{Y \sim \rho_1} [f(Y, N_0, N_1)] - \mathbb{E}_{Y \sim \rho_2} [f(Y, N_0, N_1)] \right)^2 \\ &= \left(\sum_{l=0}^N c(N_0, N_1, l) \mathbb{E}_{Y \sim \rho_1} [Y^l] - \sum_{l=0}^N c(N_0, N_1, l) \mathbb{E}_{Y \sim \rho_2} [Y^l] \right)^2 \\ &= \left(\sum_{l=0}^N c(N_0, N_1, l) \left(\mathbb{E}_{Y \sim \rho_1} [Y^l] - \mathbb{E}_{Y \sim \rho_2} [Y^l] \right) \right)^2 \\ &= \sum_{l=0}^N \sum_{l'=0}^N c(N_0, N_1, l) c(N_0, N_1, l') \left(\mathbb{E}_{Y \sim \rho_1} [Y^l] - \mathbb{E}_{Y \sim \rho_2} [Y^l] \right) \left(\mathbb{E}_{Y \sim \rho_1} [Y^{l'}] - \mathbb{E}_{Y \sim \rho_2} [Y^{l'}] \right). \end{aligned}$$

For the ease of notation, let $m_l := \mathbb{E}_{Y \sim \rho_1} [Y^l] - \mathbb{E}_{Y \sim \rho_2} [Y^l]$. Then, we obtain that

$$\left(\mathbb{E}_{Y \sim \rho_1} [f(Y, N_0, N_1)] - \mathbb{E}_{Y \sim \rho_2} [f(Y, N_0, N_1)] \right)^2 = \sum_{l=0}^N \sum_{l'=0}^N c(N_0, N_1, l) c(N_0, N_1, l') m_l m_{l'}.$$

Then, it holds that

$$\begin{aligned} & \mathbb{E}_{N_0 \sim B(N, \frac{1}{2})} \left[\sum_{l=0}^N \sum_{l'=0}^N c(N_0, N_1, l) c(N_0, N_1, l') m_l m_{l'} \right] \\ &= \sum_{N_0=0}^N \binom{N}{N_0} \left(\frac{1}{2} \right)^N \sum_{l=0}^N \sum_{l'=0}^N c(N_0, N_1, l) c(N_0, N_1, l') m_l m_{l'} \\ &= \left(\frac{1}{2} \right)^N \sum_{l=0}^N \sum_{l'=0}^N m_l m_{l'} \sum_{N_0=0}^N \binom{N}{N_0} c(N_0, N_1, l) c(N_0, N_1, l'). \end{aligned}$$

We need the following lemma, whose proof is provided at the end of this subsection.

Lemma 24. *For nonnegative integers l, l' , and N , we have the following equality:*

$$\sum_{a=0}^N \binom{N}{a} c(a, N-a, l) c(a, N-a, l') = \begin{cases} 2^N \binom{N}{l} & (l = l') \\ 0 & (l \neq l') \end{cases}. \quad (14)$$

By this lemma, we conclude that

$$\mathbb{E}_{N_0, N_1} \left[\sum_{l=0}^N \sum_{l'=0}^N c(N_0, N_1, l) c(N_0, N_1, l') m_l m_{l'} \right] = \sum_{l=0}^N \binom{N}{l} m_l^2.$$

$\square$

Proof of Lemma 24. We prove the lemma using generating functions. Consider the polynomial $((1-x)(1-y) + (1+x)(1+y))^N$. First, we have

$$\begin{aligned}
 ((1-x)(1-y) + (1+x)(1+y))^N &= \sum_{a=0}^N \binom{N}{a} ((1-x)(1-y))^a ((1+x)(1+y))^{N-a} \\
 &= \sum_{a=0}^N \binom{N}{a} ((1-x)^a (1+x)^{N-a}) ((1-y)^a (1+y)^{N-a}) \\
 &= \sum_{a=0}^N \binom{N}{a} \left(\sum_{l=0}^N c(a, N-a, l) x^l \right) \left(\sum_{l'=0}^N c(a, N-a, l') y^{l'} \right) \\
 &= \sum_{l=0}^N \sum_{l'=0}^N \sum_{a=0}^N \binom{N}{a} c(a, N-a, l) c(a, N-a, l') x^l y^{l'}.
 \end{aligned}$$

Hence, the left-hand side of Eq. (14) is the coefficient of $x^l y^{l'}$ in $((1-x)(1-y) + (1+x)(1+y))^N$. On the other hand, noting that $(1-x)(1-y) + (1+x)(1+y) = 2 + 2xy$, we have

$$\begin{aligned}
 ((1-x)(1-y) + (1+x)(1+y))^N &= (2(1+xy))^N \\
 &= 2^N \sum_{l=0}^N \binom{N}{l} (xy)^l.
 \end{aligned}$$

In this expression, the coefficient of $x^l y^{l'}$ is the right-hand side of Eq. (14), proving the equality between the two. $\square$

F.4. Proof of Lemma 23

Proof of Lemma 23. Expanding the expectation using the pmf of binomial distribution, we derive that

$$\begin{aligned}
 \mathbb{E}_{N \sim B(n,p)} \left[a^N \sum_{l=k+1}^N \binom{N}{l} x^l \right] &= \sum_{N=0}^n \binom{n}{N} p^N (1-p)^{n-N} a^N \sum_{l=k+1}^N \binom{N}{l} x^l \\
 &= \sum_{l=k+1}^n \left(\sum_{N=l}^n \binom{n}{N} \binom{N}{l} (ap)^N (1-p)^{n-N} \right) x^l.
 \end{aligned} \tag{15}$$

Consider a polynomial $g(x) := (ap(1+x) + 1-p)^n$. We have

$$\begin{aligned}
 (ap(1+x) + 1-p)^n &= \sum_{N=0}^n \binom{n}{N} (ap)^N (1+x)^N (1-p)^{n-N} \\
 &= \sum_{N=0}^n \binom{n}{N} (ap)^N (1-p)^{n-N} \sum_{l=0}^N \binom{N}{l} x^l \\
 &= \sum_{l=0}^n \left(\sum_{N=l}^n \binom{n}{N} \binom{N}{l} (ap)^N (1-p)^{n-N} \right) x^l.
 \end{aligned}$$

Observe that Eq. (15) is a remainder of $g(x)$ when approximated by a degree k polynomial. By Taylor's theorem, for fixed x , there exists $\tau \in [0, 1]$ such that

$$\begin{aligned}
 \sum_{l=k+1}^n \left(\sum_{N=l}^n \binom{n}{N} \binom{N}{l} (ap)^N (1-p)^{n-N} \right) x^l &= g(x) - \sum_{l=0}^k \left(\sum_{N=l}^n \binom{n}{N} \binom{N}{l} (ap)^N (1-p)^{n-N} \right) x^l \\
 &= \frac{x^{k+1}}{(k+1)!} g^{(k+1)}(\tau x).
 \end{aligned}$$

The proof is completed by noting that $\frac{1}{(k+1)!} g^{(k+1)}(x) = \binom{n}{k+1} (ap)^{k+1} (ap(1+x) + 1-p)^{n-k-1}$. $\square$

G. Auxiliary Lemmas

Lemma 25 (Bennett's inequality, Theorem 3 in [Maurer & Pontil \(2009\)](#)). *For $n \geq 1$, let $Z_1, \dots, Z_n$ be i.i.d. copies of a random variable Z that lies in $[0, 1]$. Let $\bar{Z}_n := \frac{1}{n} \sum_{i=1}^n Z_i$ and $V := \mathbb{E}[(Z - \mathbb{E}[Z])^2]$. Then, for $\delta \in (0, 1]$, it holds that with probability at least $1 - \delta$,*

$$\mathbb{E}[Z] - \bar{Z}_n \leq \sqrt{\frac{2V_n \log \frac{1}{\delta}}{n}} + \frac{\log \frac{1}{\delta}}{3n}.$$

Lemma 26 (Theorem 4 in [Maurer & Pontil \(2009\)](#)). *For $n \geq 2$, let $Z_1, \dots, Z_n$ be i.i.d. copies of a random variable Z that lies in $[0, 1]$. Let $\bar{Z}_n := \frac{1}{n} \sum_{i=1}^n Z_i$ and $\hat{V}_n := \frac{1}{n} \sum_{i=1}^n (Z_i - \bar{Z}_n)^2$. Then, for $\delta \in (0, 1]$, it holds that with probability at least $1 - \delta$,*

$$\mathbb{E}[Z] - \bar{Z}_n \leq 2\sqrt{\frac{\hat{V}_n \log \frac{2}{\delta}}{n}} + \frac{14 \log \frac{2}{\delta}}{3n}.$$

Lemma 27 (Hoeffding's inequality, Theorem 2 in [Hoeffding \(1963\)](#)). *Let $\{\xi_i\}_{i=1}^n$ be a sequence of real-valued random variables adapted to a filtration $\{\mathcal{F}_i\}_{i=0}^n$. Suppose that there exist $a_i < b_i$ such that $\xi_i \in [a_i, b_i]$ holds almost surely for all $i \in [n]$. Then, for $\delta \in (0, 1]$, the following inequality holds with probability at least $1 - \delta$:*

$$\sum_{i=1}^n (\xi_i - \mathbb{E}[\xi_i | \mathcal{F}_{t-1}]) \leq \sqrt{\frac{1}{2} \left(\sum_{i=1}^n (b_i - a_i)^2 \right) \log \frac{1}{\delta}}.$$

Lemma 28 (Pinsker's inequality, Lemma 2.5 in [Tsybakov \(2008\)](#)). *For two probability measures P and Q , we have $V(P, Q) \leq \sqrt{\frac{1}{2} \text{KL}(P \parallel Q)}$, where $V(P, Q)$ is the total variation of the two measures.*